Lecture: Time Response

- Transient response measurement (TRM)
  - First order system and their TRM parameters
  - System identifications G(s)
  - Second order system and their TRM parameters
  - Classification of second order system
  - Alternative Representation of Second order system
  - Design Problem R, C, L K, J, D
  - System with additional Poles

Lecture: Time Response

$$G(s) = \frac{s+2}{s^2+2s+5}$$

$n = 2$

A

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}_{2\times 2} \quad 2$$

$$\det(sI - A) = (s-1)(s-5)$$

2

- Transient response measurement
  - First order system
  - Second order system

- Determining order of the systems
  - Maximum power of s in denominator polynomial
  - Maximum power of s in det(sI-A) polynomial
  - Number of storage elements in an electrical or mechanical system
  - Highest order of derivative in a differential equations presenting a system.

$$\frac{d^2y}{dt^2} + 2y = 5u$$

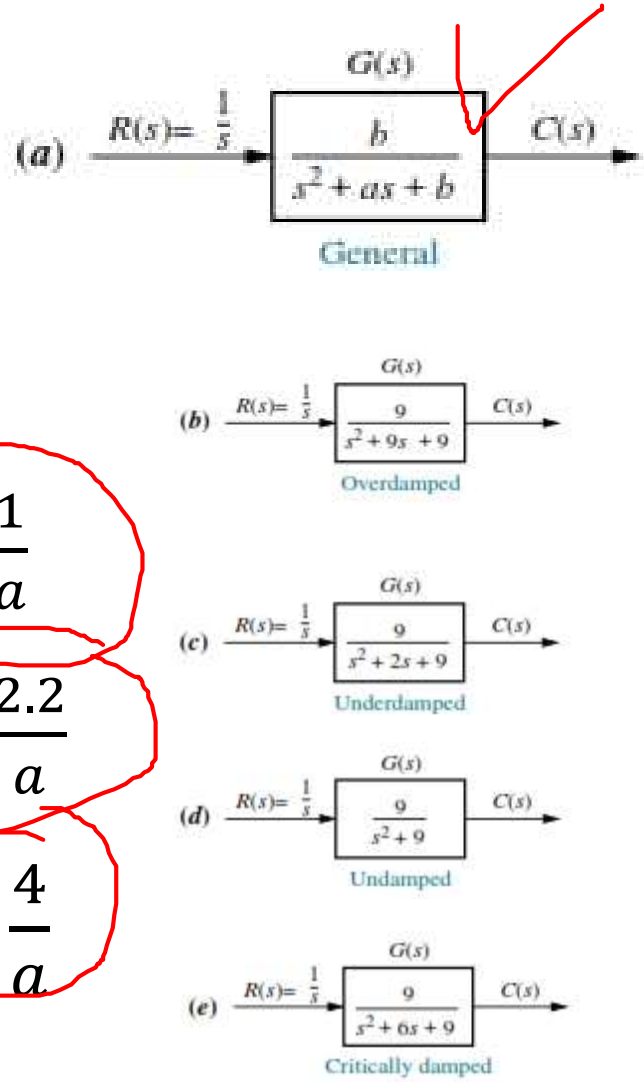
Lecture: Time Response

$$G(s) = \frac{a}{s+a}$$

Time constant, $T_c = \frac{1}{a}$

Rise Time, $T_r = \frac{2.2}{a}$

Settling Time, $T_s = \frac{4}{a}$



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

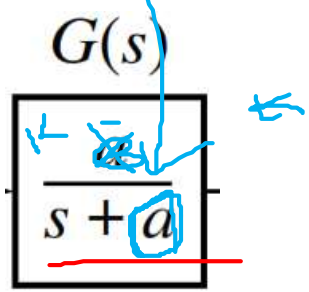
$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

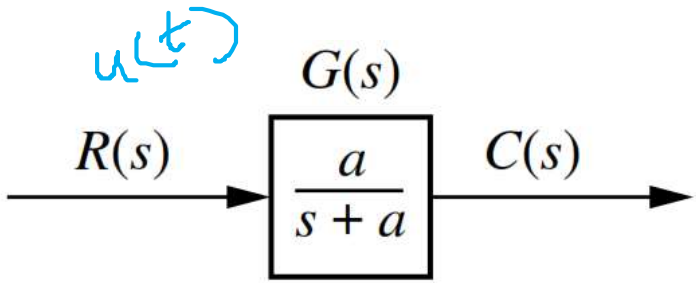
$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

$$T_s = \frac{4}{\zeta\omega_n}$$

$$T_r = \frac{2.2}{\zeta\omega_n}$$

Lecture: Time Response





$$C(s) = R(s)G(s) = \frac{a}{s(s+a)}$$

$$\frac{a}{s(s+a)} = \frac{A}{s} + \frac{B}{s+a} \quad (1)$$

$s = -a$

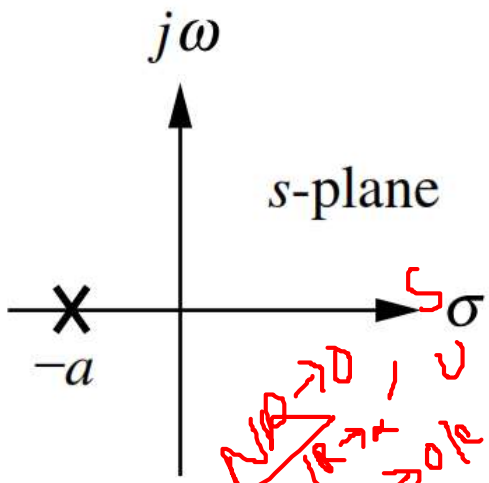
$a = 0 + B(-a)$

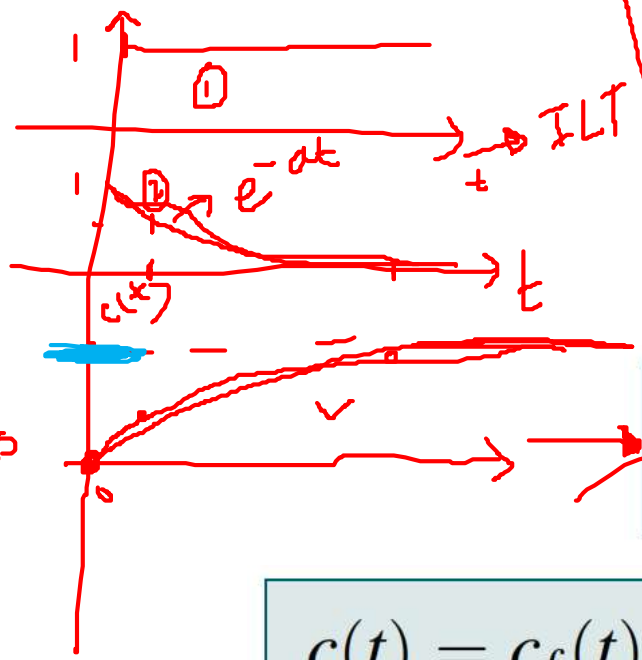
$B = -1 = -K/a$

$s = 0$

$a = AB$

$A = 1$





$$c(t) = 1 - e^{-at}$$

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at}$$

Lecture: Time Response

$c(t) = 1 - e^{-at}$ — (1)

$t = \frac{1}{a}$

$c(t) = 1 - e^{-1}$
$= 0.63 \rightarrow 63\%$ of FV

$T_c = \frac{1}{a}$ ✓

$t = T_s$

$0.98 = 1 - e^{-aT_s}$

$\Rightarrow e^{-aT_s} = 0.02$

$-aT_s = -4$

$T_s = \frac{4}{a}$

$t = T_1$

$0.1 = 1 - e^{-aT_1}$

$\Rightarrow e^{-aT_1} = 0.9$

$t = T_2$

$0.9 = 1 - e^{-aT_2}$

$\frac{e^{-aT_2}}{e^{-aT_1}} = \frac{0.1}{0.9} = 0.11$

$e^{-a(T_2 - T_1)} = 0.11$

$-aT_r = -2.2$

$T_r = \frac{2.2}{a}$ ✓

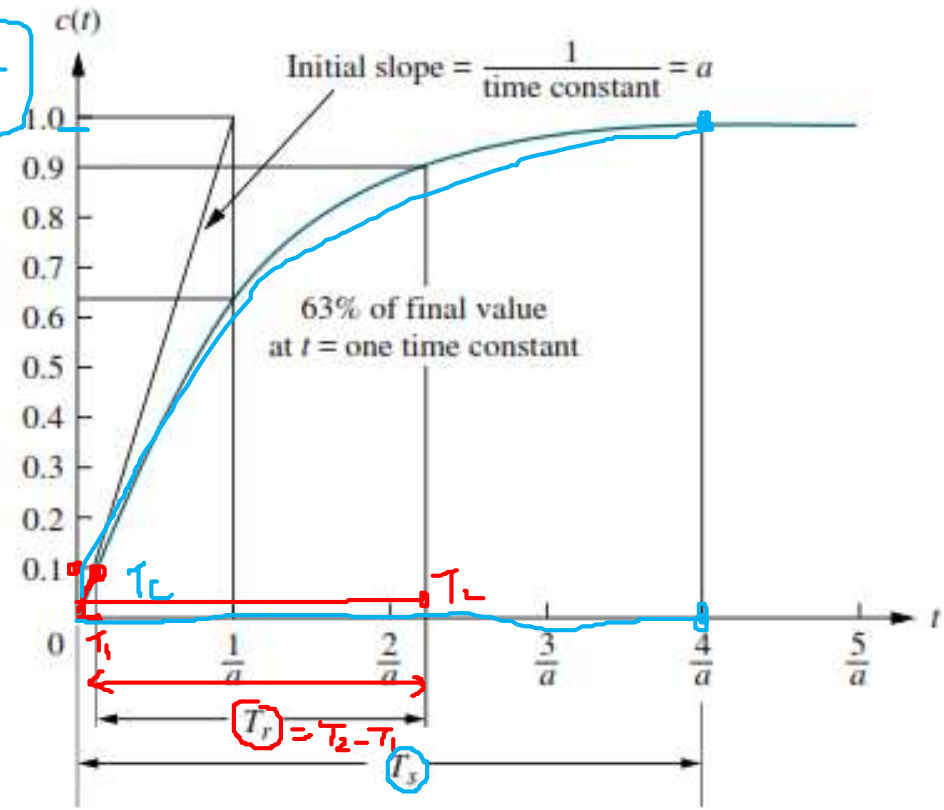


FIGURE 4.5 First-order system response to a unit step

Time Constant

We call $1/a$ the *time constant* of the response. From Eq. (4.7), the time constant can be described as the time for e^{-at} to decay to 37% of its initial value. Alternately, from Eq. (4.8) the time constant is the time it takes for the step response to rise to 63% of its final value (see Figure 4.5).

Rise Time, T_r

Rise time is defined as the time for the waveform to go from 0.1 to 0.9 of its final value. Rise time is found by solving Eq. (4.6) for the difference in time at $c(t) = 0.9$ and $c(t) = 0.1$. Hence,

$$\boxed{T_r = \frac{2.31}{a} - \frac{0.11}{a} = \frac{2.2}{a}} \tag{4.9}$$

Settling Time, T_s

Settling time is defined as the time for the response to reach, and stay within, 2% of its final value.[2] Letting $c(t) = 0.98$ in Eq. (4.6) and solving for time, t, we find the settling time to be

$$\boxed{T_s = \frac{4}{a}} \tag{4.10}$$

First-Order Transfer Functions via Testing

$$C(s) = \frac{K}{s(s+a)} = \frac{K/a}{s} - \frac{K/a}{(s+a)}$$

$a = 1/0.13 = 7.7.$

$K/a = 0.72.$

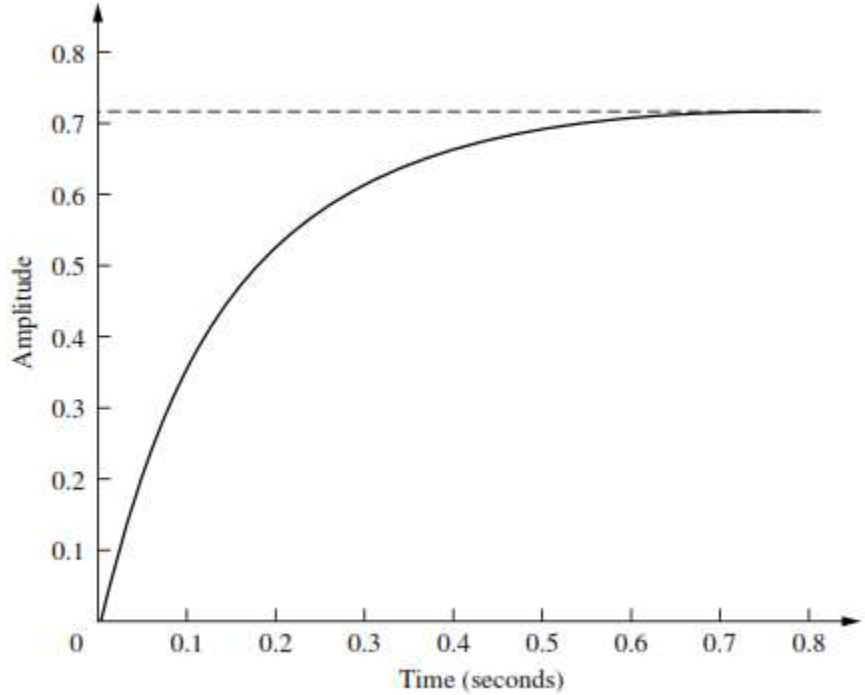
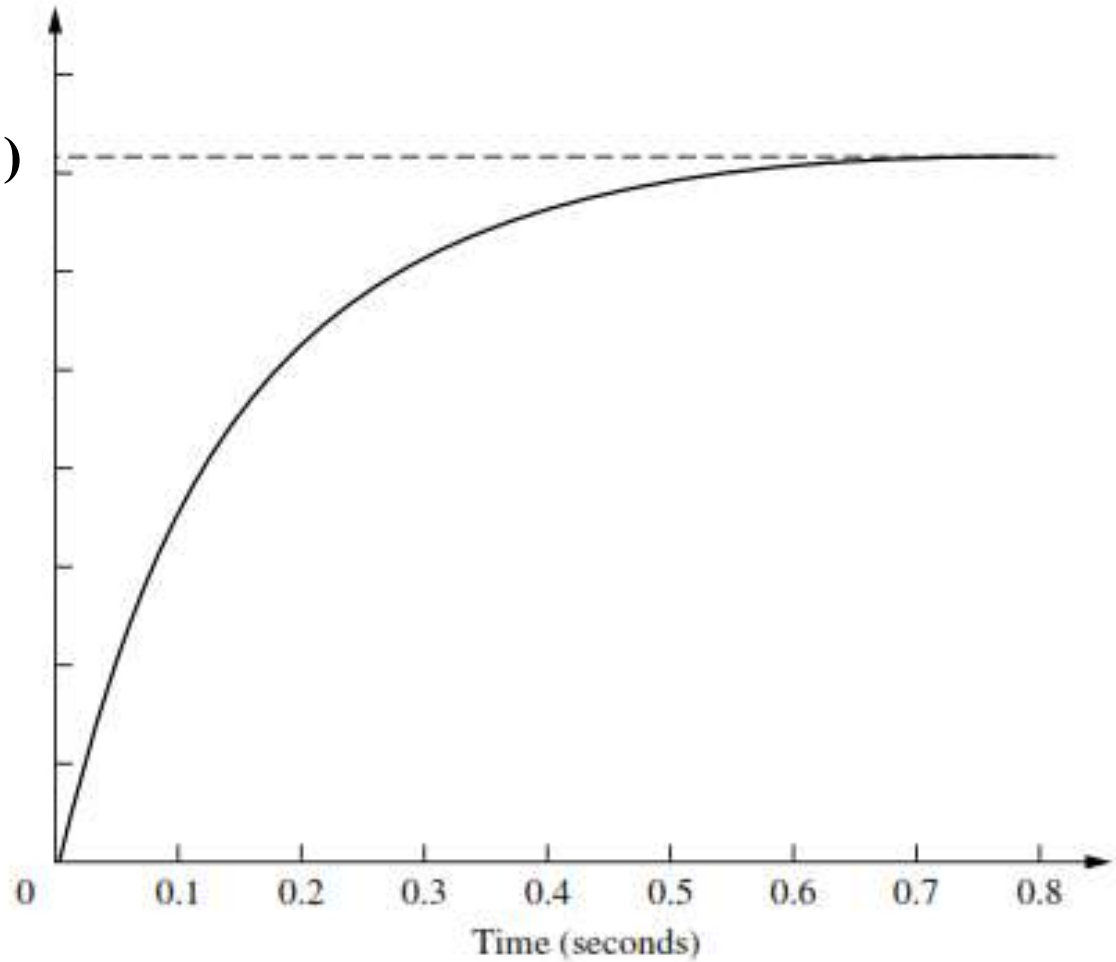


FIGURE 4.6 Laboratory results of a system step response test

(ID last digit+1)



Laboratory results of a system step response test

Answer the following:

a) What is the order of the system?
b) Determine the transfer function of the system.

First-Order Transfer Functions via Testing

Skill-Assessment Exercise 4.2

PROBLEM: A system has a transfer function, $G(s) = \frac{50}{s+50}$. Find the time constant, T_c, settling time, T_s, and rise time, T_r.

ANSWER: $T_c = 0.02\,\text{s}$, $T_s = 0.08\,\text{s}$, and $T_r = 0.044\,\text{s}$.

4.3 First-Order Systems

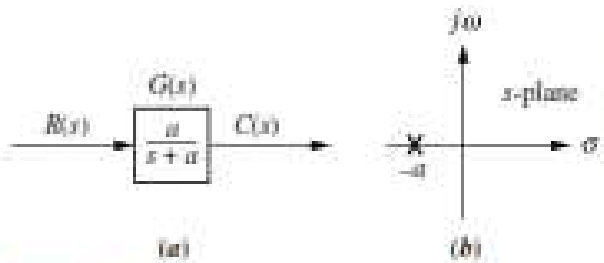


FIGURE 4.4 **a.** First-order system; **b.** pole plot

We now discuss first-order systems without zeros to define a performance specification for such a system. A first-order system without zeros can be described by the transfer function shown in Figure 4.4(*a*). If the input is a unit step, where $R(s) = 1/s$, the Laplace transform of the step response is $C(s)$, where

$$C(s) = R(s)G(s) = \frac{a}{s(s+a)} \tag{4.5}$$

Taking the inverse transform, the step response is given by

$$c(t) = c_f(t) + c_n(t) = 1 - e^{-at} \tag{4.6}$$

Virtual Experiment 4.1
First-Order
Open-Loop Systems

Put theory into practice and find a first-order transfer function representing the Quanser Rotary Servo. Then validate the model by simulating it in LabVIEW. Such a servo motor is used in

where the input pole at the origin generated the forced response $c_f(t) = 1$, and the system pole at $-a$, as shown in Figure 4.4(*b*), generated the natural response $c_n(t) = -e^{-at}$. Equation (4.6) is plotted in Figure 4.5.

Let us examine the significance of parameter a, the only parameter needed to describe the transient response. When $t = 1/a$,

$$e^{-at}|_{t=1/a} = e^{-1} = 0.37 \tag{4.7}$$

or

$$c(t)|_{t=1/a} = 1 - e^{-at}|_{t=1/a} = 1 - 0.37 = 0.63 \tag{4.8}$$

[redacted]

4.4 S

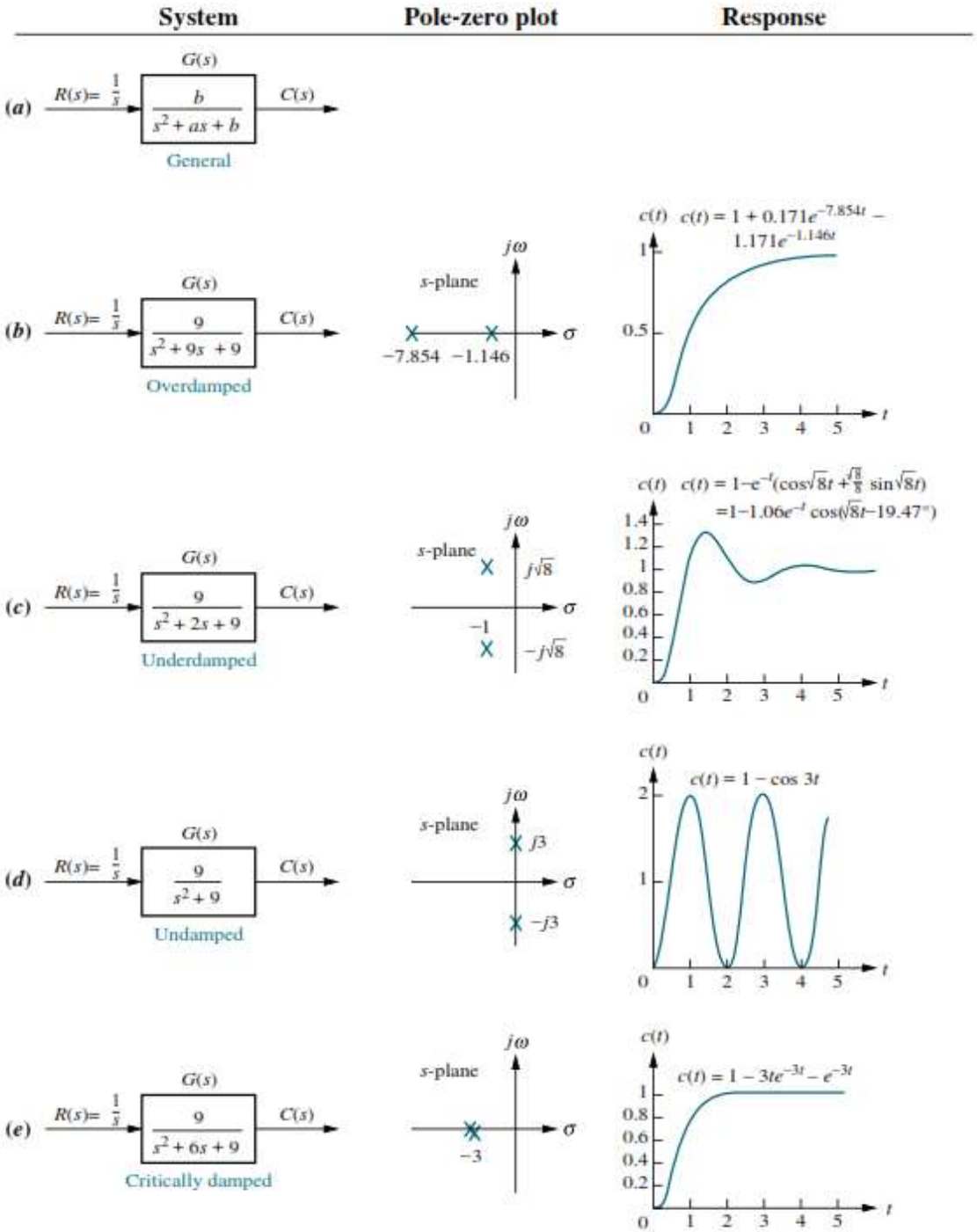


FIGURE 4.7 Second-order systems, pole plots, and step responses

[redacted]

4.4 Second-Order Systems: Introduction

1. *Overdamped responses*

   Poles: Two real at $-\sigma_1,\ -\sigma_2$

   Natural response: Two exponentials with time constants equal to the reciprocal of the pole locations, or

$$c(t) = K_1 e^{-\sigma_1 t} + K_2 e^{-\sigma_2 t}$$

2. *Underdamped responses*

   Poles: Two complex at $-\sigma_d \pm j\omega_d$

   Natural response: Damped sinusoid with an exponential envelope whose time constant is equal to the reciprocal of the pole's real part. The radian frequency of the sinusoid, the damped frequency of oscillation, is equal to the imaginary part of the poles, or

$$c(t) = Ae^{-\sigma_d t}\cos(\omega_d t - \phi)$$

3. *Undamped responses*

   Poles: Two imaginary at $\pm j\omega_1$

   Natural response: Undamped sinusoid with radian frequency equal to the imaginary part of the poles, or

$$c(t) = \mathrm{A}\cos(\omega_1 t - \phi)$$

4. *Critically damped responses*

   Poles: Two real at $-\sigma_1$

$$c(t) = K_1 e^{-\sigma_1 t} + K_2 t e^{-\sigma_1 t}$$

4.4 Second-Order Systems: Introduction

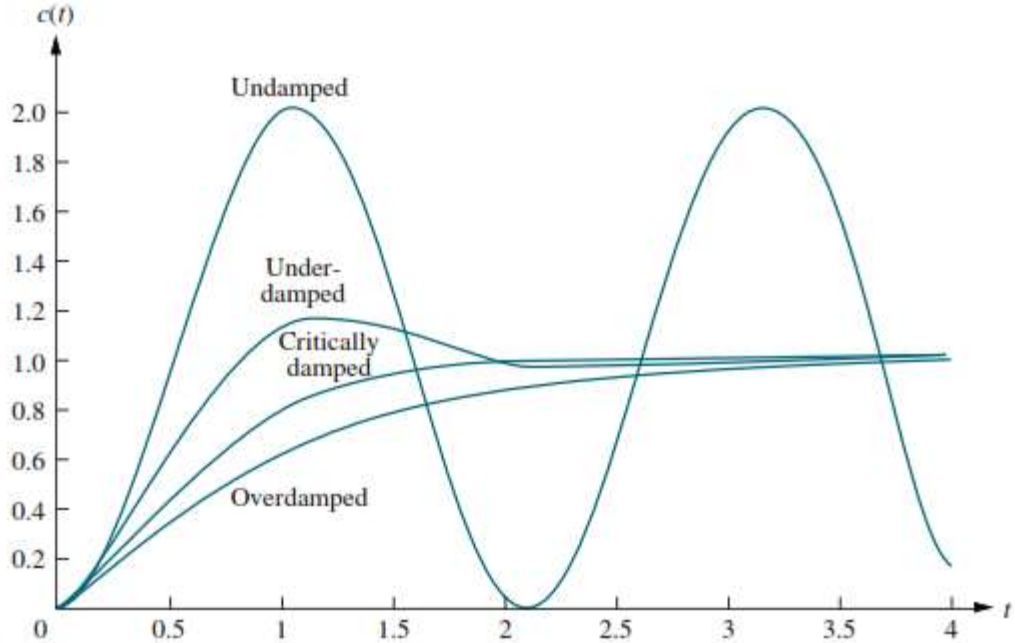


FIGURE 4.10 Step responses for second-order system damping cases

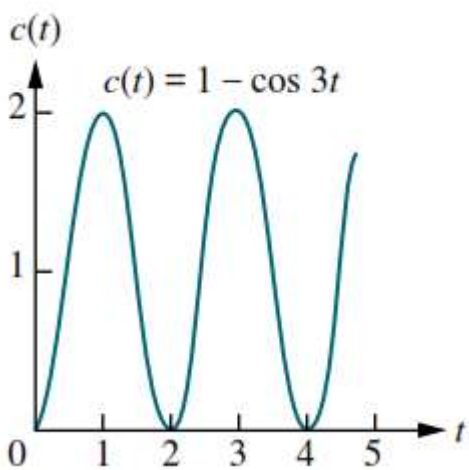
4.5 The General Second-Order System

Natural Frequency, ω_n

The *natural frequency* of a second-order system is the frequency of oscillation of the system without damping. For example, the frequency of oscillation of a series *RLC* circuit with the resistance shorted would be the natural frequency.

Damping Ratio, ζ

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency (rad/second)}} = \frac{1}{2\pi}\,\frac{\text{Natural period (seconds)}}{\text{Exponential time constant}}$$



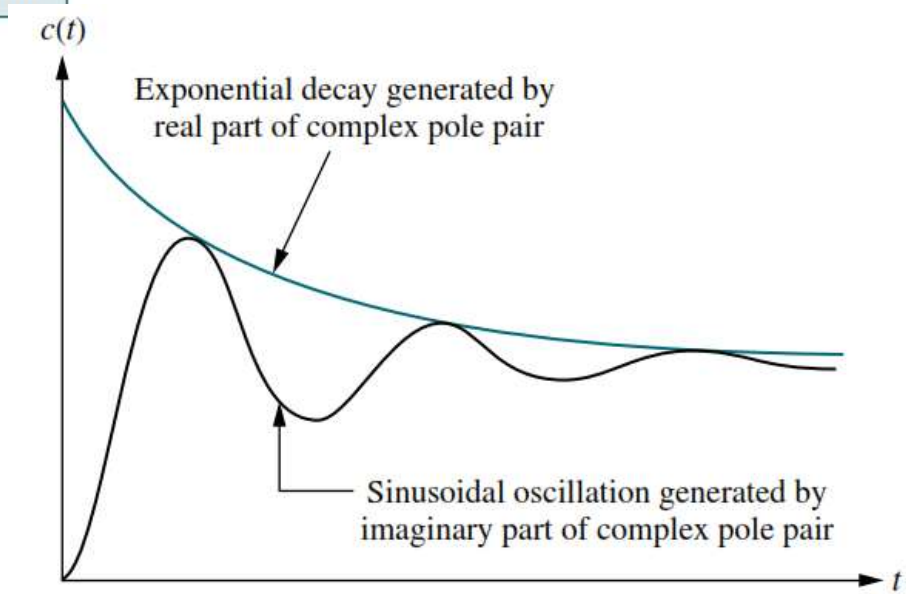


FIGURE 4.8 Second-order step response components generated by complex poles

4.5 The General Second-Order System

$$G(s) = \frac{b}{s^2 + as + b}$$

For the poles to be purely imaginary, $a = 0$. Hence,

$$G(s) = \frac{b}{s^2 + b}$$

By definition, the natural frequency, ω_n, is the frequency of oscillation of this system. Since the poles of this system are on the $j\omega$-axis at $\pm j\sqrt{b}$,

$$\omega_n = \sqrt{b} \tag{4.18}$$

Hence,

$$b = \omega_n^2 \tag{4.19}$$

$$\zeta = \frac{\text{Exponential decay frequency}}{\text{Natural frequency (rad/second)}} = \frac{|\sigma|}{\omega_n} = \frac{a/2}{\omega_n}$$

$$a = 2\zeta\omega_n$$

$$\boxed{G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

4.5 The General Second-Order System

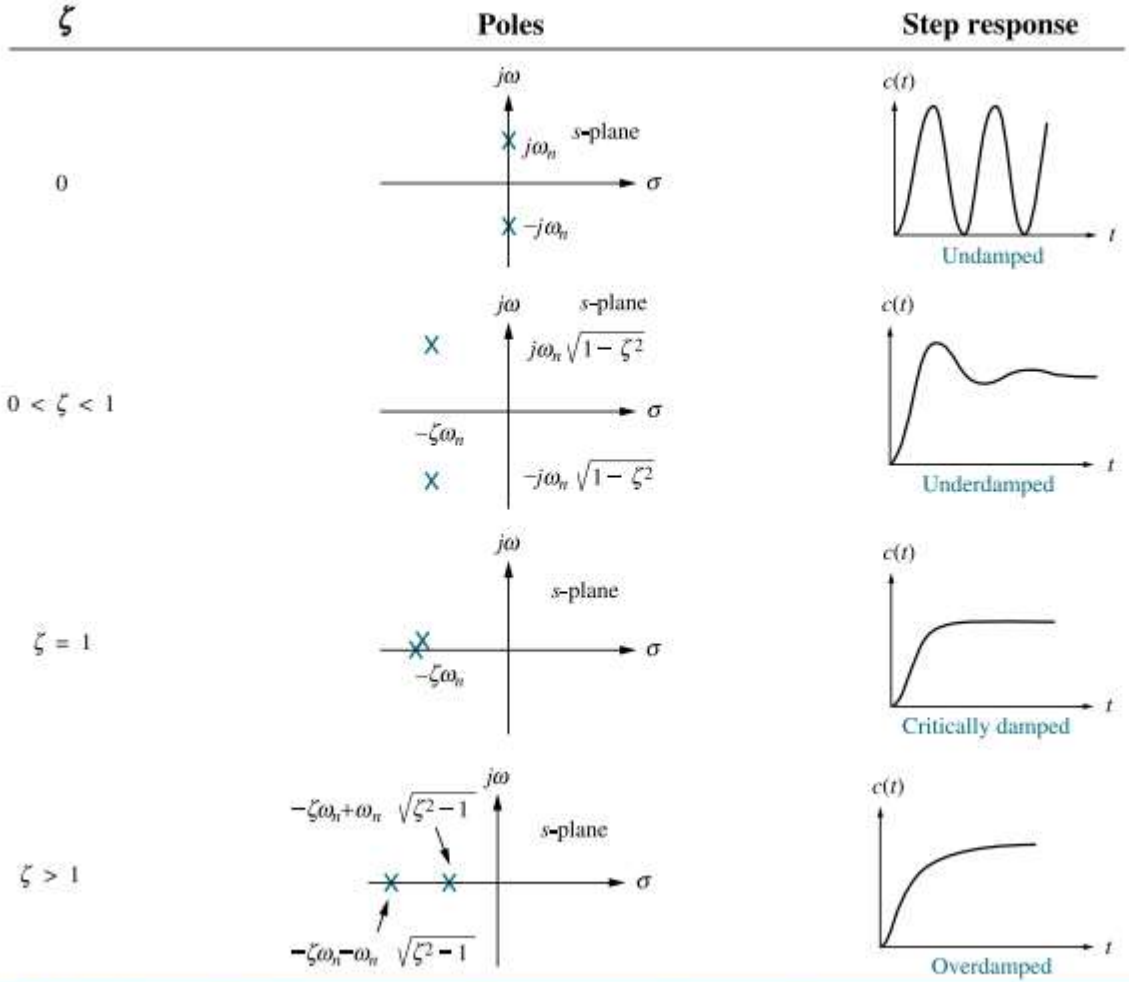


FIGURE 4.11 Second-order response as a function of damping ratio

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

4.5 The General Second-Order System

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Example 4.4

Characterizing Response from the Value of ζ

PROBLEM: For each of the systems shown in Figure 4.12, find the value of ζ and report the kind of response expected.

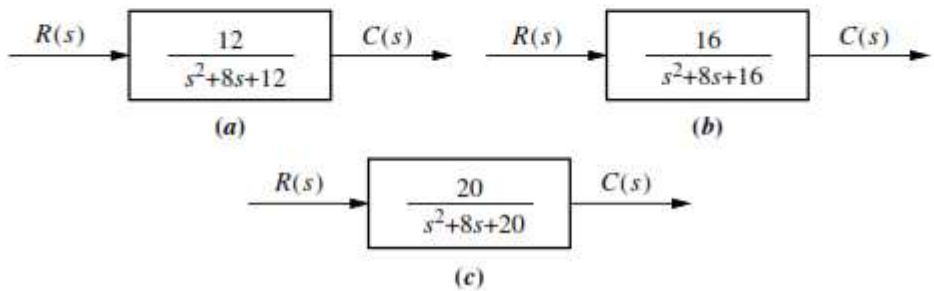


FIGURE 4.12 Systems for Example 4.4

4.6 Underdamped Second-Order Systems

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta}{\sqrt{1-\zeta^2}}\omega_n\sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + \omega_n^2(1-\zeta^2)}$$

$$\begin{aligned} c(t) &= 1 - e^{-\zeta\omega_n t}\left(\cos\omega_n\sqrt{1-\zeta^2}t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_n\sqrt{1-\zeta^2}t\right) \\ &= 1 - \frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\cos(\omega_n\sqrt{1-\zeta^2}t - \phi) \end{aligned} \tag{4.28}$$

where $\phi = \tan^{-1}(\zeta/\sqrt{1-\zeta^2})$.

4.6 Underdamped Second-Order Systems

1. *Rise time,* T_r. The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.
2. *Peak time,* T_P. The time required to reach the first, or maximum, peak.
3. *Percent overshoot,* $\%OS$. The amount that the waveform overshoots the steady-state, or final, value at the peak time, expressed as a percentage of the steady-state value.
4. *Settling time,* T_s. The time required for the transient's damped oscillations to reach and stay within ±2% of the steady-state value.

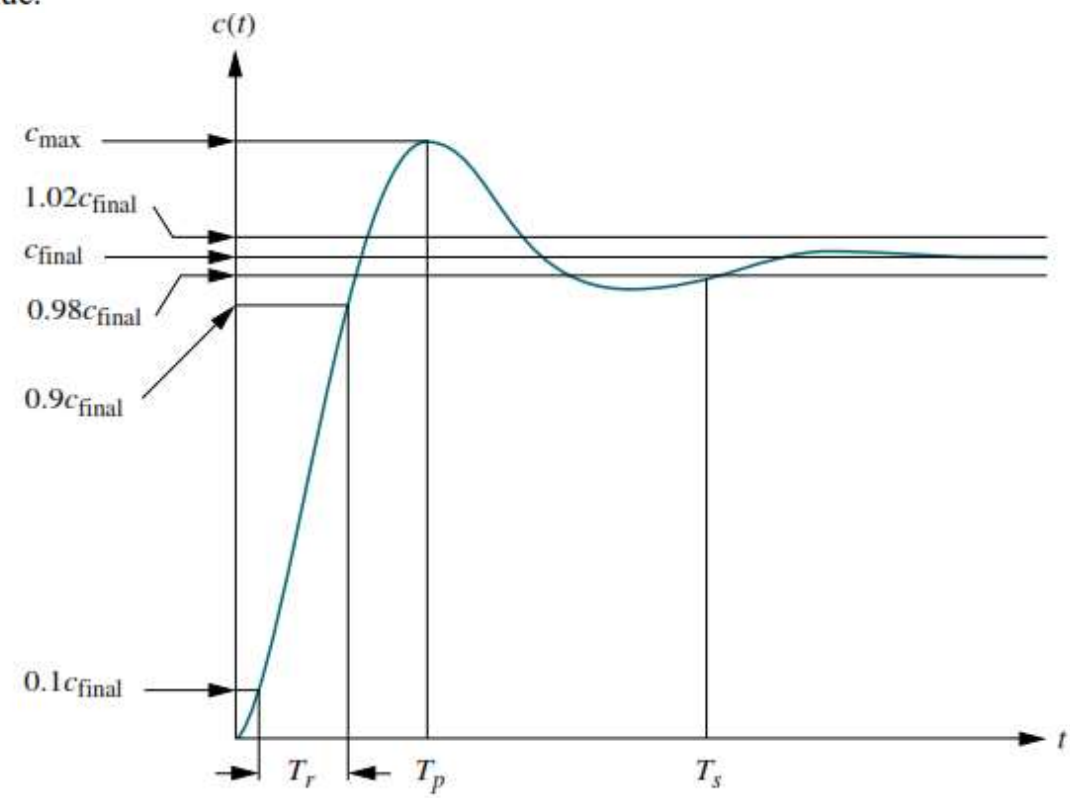


FIGURE 4.14 Second-order underdamped response specifications

4.6 Underdamped Second-Order Systems

1. *Rise time, T_r.* The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.
2. *Peak time, T_P.* The time required to reach the first, or maximum, peak.
3. *Percent overshoot, $\%OS$.* The amount that the waveform overshoots the steady-state, or final, value at the peak time, expressed as a percentage of the steady-state value.
4. *Settling time, T_s.* The time required for the transient's damped oscillations to reach and stay within ±2% of the steady-state value.

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

$$T_s = \frac{4}{\zeta\omega_n}$$

4.6 Underdamped Second-Order Systems

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

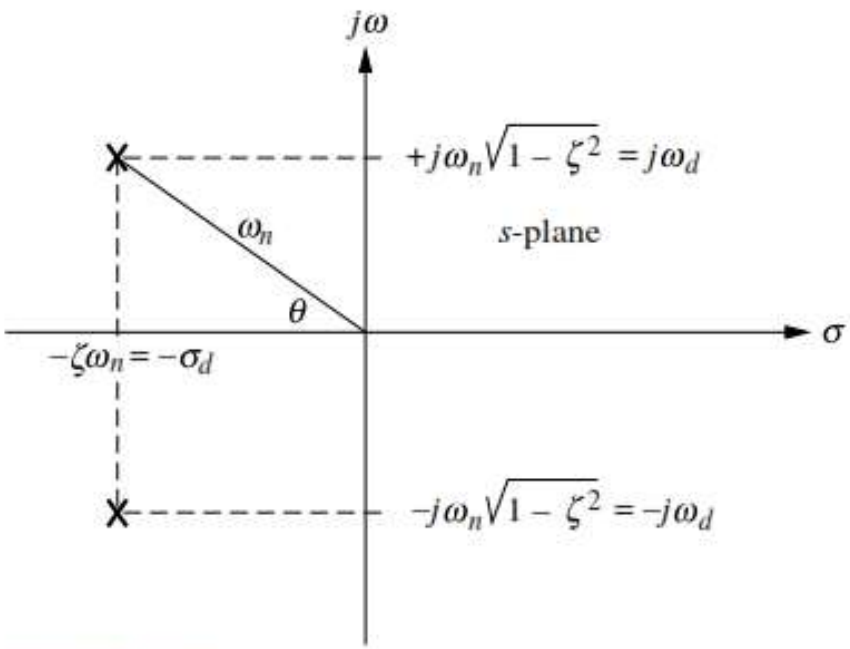


FIGURE 4.17 Pole plot for an underdamped

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{\pi}{\sigma_d}$$

where ω_d is the imaginary part of the pole and is called the *damped frequency of oscillation*, and σ_d is the magnitude of the real part of the pole and is the *exponential damping frequency*.

4.6 Underdamped Second-Order Systems

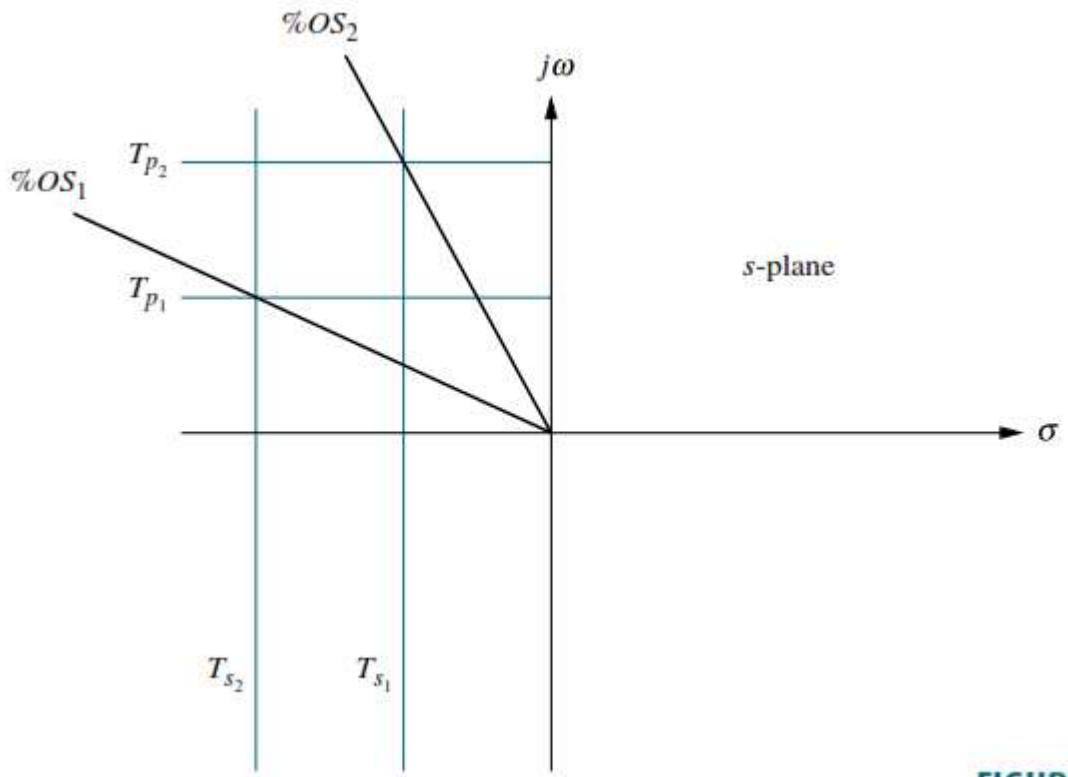


FIGURE 4.18 Lines of constant peak time, T_p, settling time, T_s, and percent overshoot, $\%OS$. Note: $T_{s_2} < T_{s_1}$; $T_{p2} < T_{p1}$; $\%OS_1 < \%OS_2$.

4.6 Underdamped

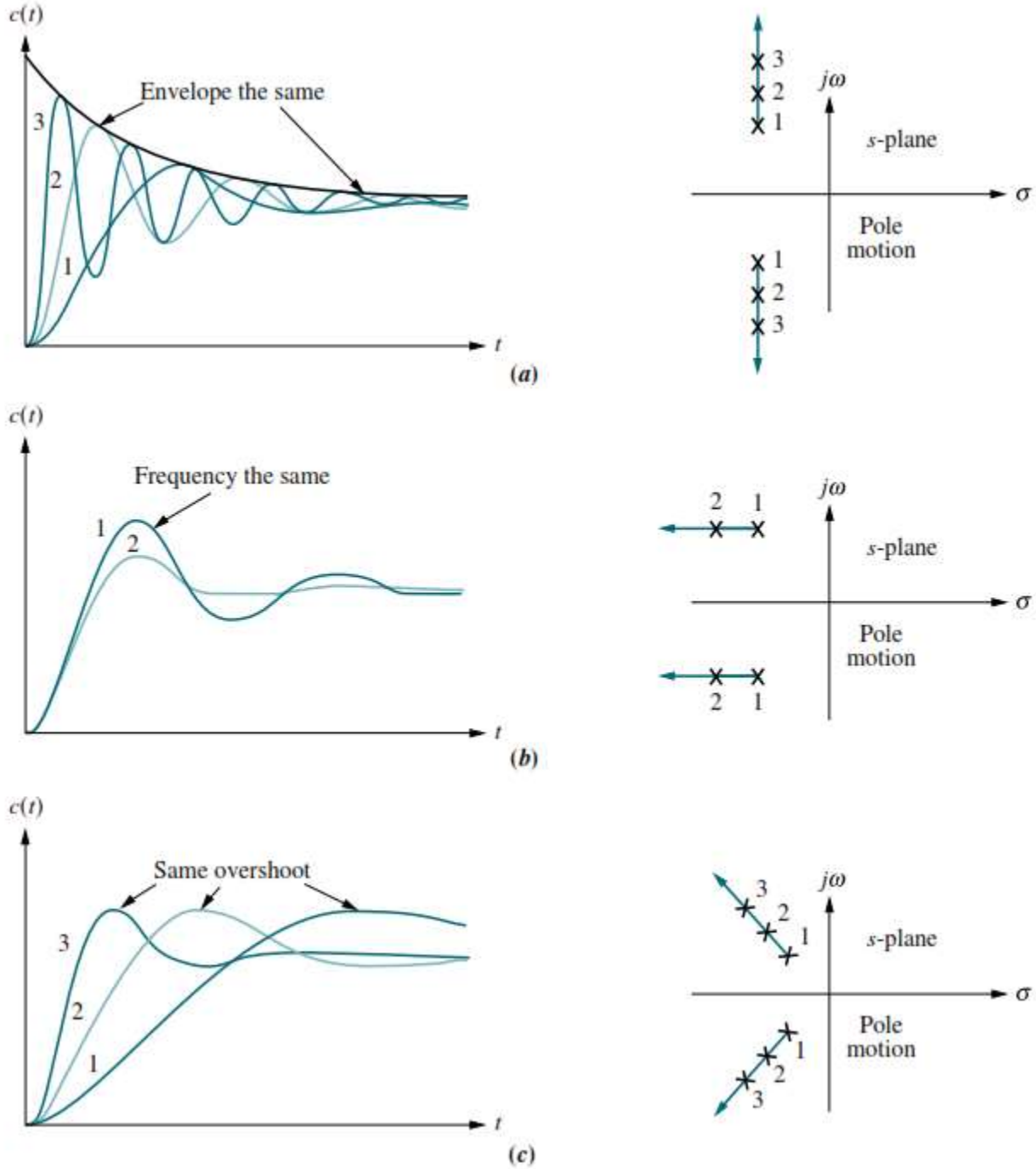


FIGURE 4.19 Step responses of second-order underdamped systems as poles move: **a.** with constant real part; **b.** with constant imaginary part; **c.** with constant damping ratio

4.6 Underdamped Second-Order Systems

Example 4.7

Transient Response Through Component Design

Design D

PROBLEM: Given the system shown in Figure 4.21, find J and D to yield 20% overshoot and a settling time of 2 seconds for a step input of torque $T(t)$.

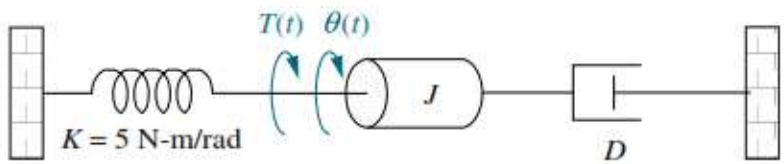


FIGURE 4.21 Rotational mechanical system for Example 4.7

SOLUTION: First, the transfer function for the system is

$$G(s) = \frac{1/J}{s^2 + \frac{D}{J}s + \frac{K}{J}} \tag{4.49}$$

From the transfer function,

$$\omega_n = \sqrt{\frac{K}{J}} \tag{4.50}$$

and

$$2\zeta\omega_n = \frac{D}{J} \tag{4.51}$$

4.6

PROBLEM:
overshoot a

FIGURE 4.21

Design
D

SOLUTION: First, the transfer function for the system is

$$G(s) = \frac{1/J}{s^2 + \frac{D}{J}s + \frac{K}{J}} \tag{4.49}$$

From the transfer function,

$$\omega_n = \sqrt{\frac{K}{J}} \tag{4.50}$$

and

$$2\zeta\omega_n = \frac{D}{J} \tag{4.51}$$

But, from the problem statement,

$$T_s = 2 = \frac{4}{\zeta\omega_n} \tag{4.52}$$

or $\zeta\omega_n = 2$. Hence,

$$2\zeta\omega_n = 4 = \frac{D}{J} \tag{4.53}$$

Also, from Eqs. (4.50) and (4.52),

$$\zeta = \frac{4}{2\omega_n} = 2\sqrt{\frac{J}{K}} \tag{4.54}$$

From Eq. (4.39), a 20% overshoot implies $\zeta = 0.456$. Therefore, from Eq. (4.54),

$$\zeta = 2\sqrt{\frac{J}{K}} = 0.456 \tag{4.55}$$

4.6 Underdamped Second-Order Systems

Hence,

$$\frac{J}{K} = 0.052 \tag{4.56}$$

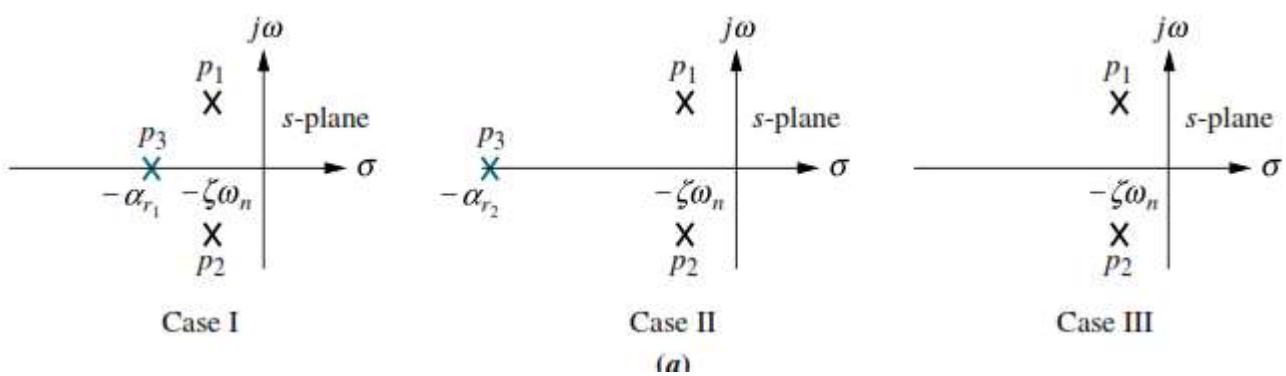
From the problem statement, $K = 5$ N-m/rad. Combining this value with Eqs. (4.53) and (4.56), $D = 1.04$ N-m-s/rad, and $J = 0.26$ kg-m^2.

4.6 Underdamped Second-Order Systems

4.7 System Response with Additional Poles

$$C(s) = \frac{A}{s} + \frac{B(s+\zeta\omega_n) + C\omega_d}{(s+\zeta\omega_n)^2 + \omega_d^2} + \frac{D}{s+\alpha_r}$$

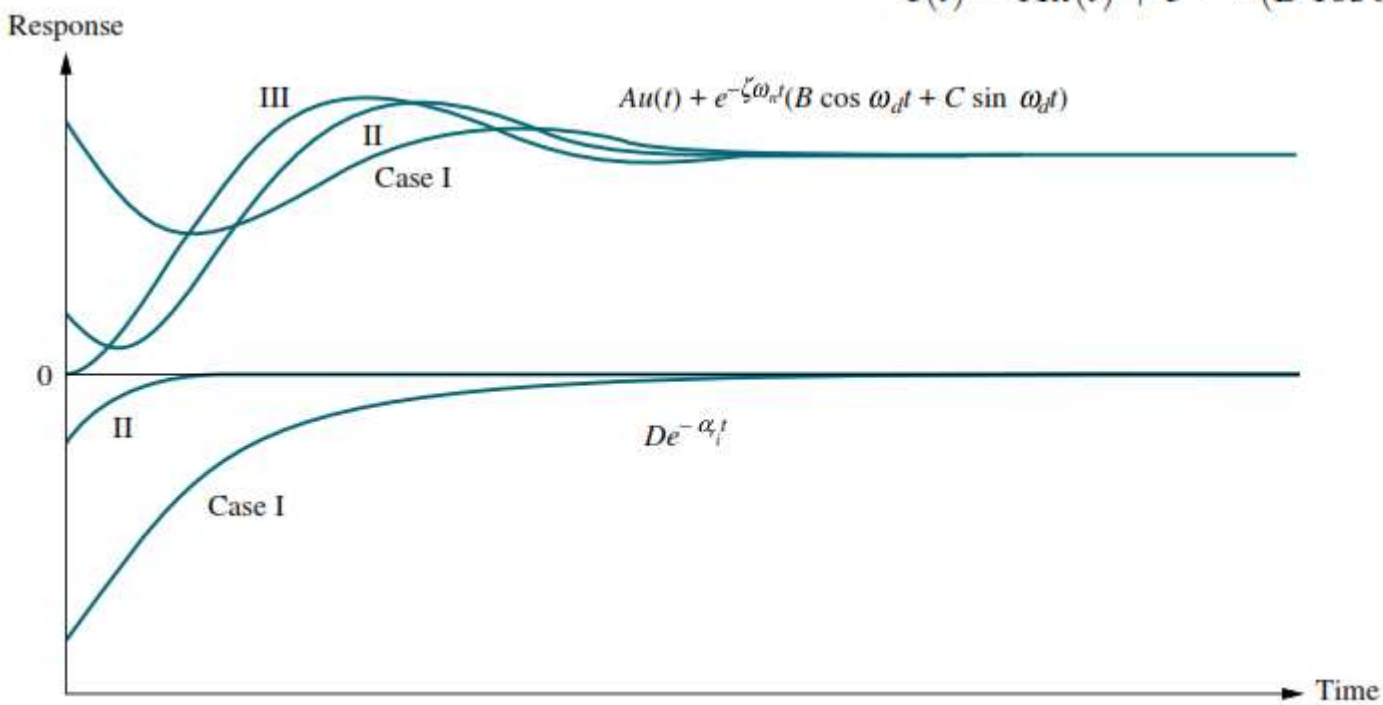
$$c(t) = Au(t) + e^{-\zeta\omega_n t}(B\cos\omega_d t + C\sin\omega_d t) + De^{-\alpha_r t}$$



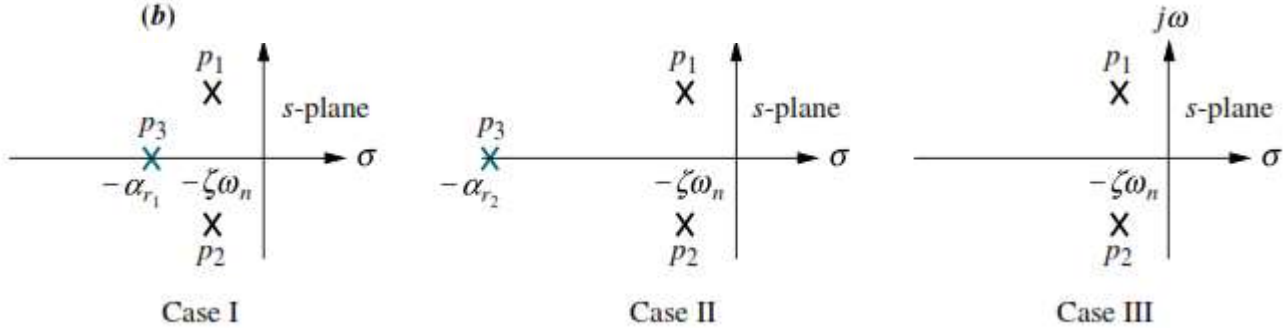
(a)

4.7 System Response with Additional Poles

$$c(t) = Au(t) + e^{-\zeta\omega_n t}(B\cos\omega_d t + C\sin\omega_d t) + De^{-\alpha_r t}$$



(b)



(a)

4.7 System Response with Additional Poles

$$c(t) = Au(t) + e^{-\zeta\omega_n t}(B\cos\omega_d t + C\sin\omega_d t) + De^{-\alpha_r t}$$

$$C(s) = \frac{bc}{s(s^2 + as + b)(s + c)} = \frac{A}{s} + \frac{Bs + C}{s^2 + as + b} + \frac{D}{s + c}$$

$$A = 1; \qquad B = \frac{ca - c^2}{c^2 + b - ca}$$

$$C = \frac{ca^2 - c^2a - bc}{c^2 + b - ca}; \quad D = \frac{-b}{c^2 + b - ca}$$

As the nondominant pole approaches ∞, or $c \to \infty$,

$$A = 1;\ B = -1;\ C = -a;\ D = 0 \tag{4.61}$$

Thus, for this example, D, the residue of the nondominant pole and its response, becomes zero as the nondominant pole approaches infinity.

4.7 System Response with Additional Poles

Example 4.8

Comparing Responses of Three-Pole Systems

PROBLEM: Find the step response of each of the transfer functions shown in Eqs. (4.62) through (4.64) and compare them.

$$T_1(s) = \frac{24.542}{s^2 + 4s + 24.542} \tag{4.62}$$

$$T_2(s) = \frac{245.42}{(s + 10)(s^2 + 4s + 24.542)} \tag{4.63}$$

$$T_3(s) = \frac{73.626}{(s + 3)(s^2 + 4s + 24.542)} \tag{4.64}$$

SOLUTION: The step response, $C_i(s)$, for the transfer function, $T_i(s)$, can be found by multiplying the transfer function by $1/s$, a step input, and using partial-fraction expansion followed by the inverse Laplace transform to find the response, $c_i(t)$. With the details left as an exercise for the student, the results are

$$c_1(t) = 1 - 1.09e^{-2t}\cos(4.532t - 23.8^\circ) \tag{4.65}$$

$$c_2(t) = 1 - 0.29e^{-10t} - 1.189e^{-2t}\cos(4.532t - 53.34^\circ) \tag{4.66}$$

$$c_3(t) = 1 - 1.14e^{-3t} + 0.707e^{-2t}\cos(4.532t + 78.63^\circ) \tag{4.67}$$

The three responses are plotted in Figure 4.24. Notice that $c_2(t)$, with its third pole at -10 and farthest from the dominant poles, is the better approximation of $c_1(t)$,

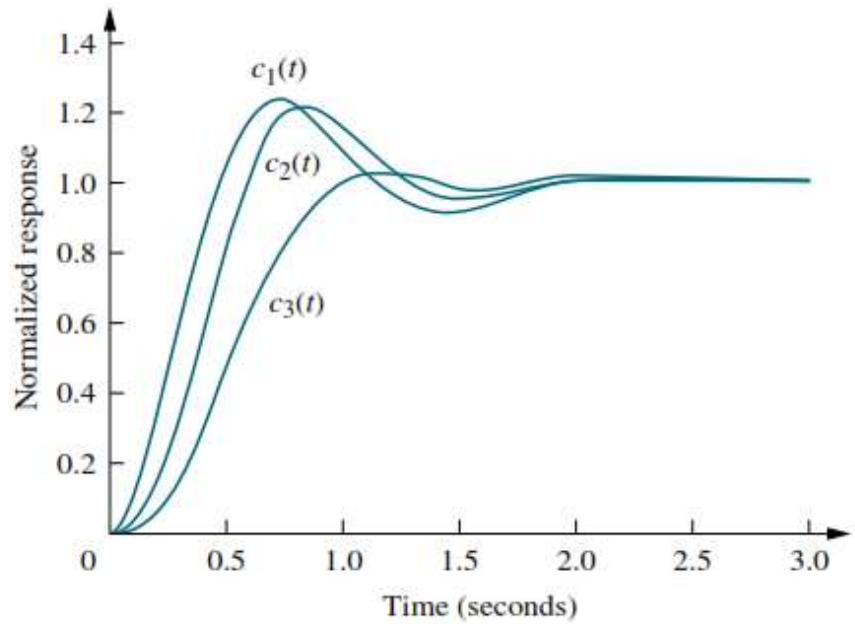


FIGURE 4.24 Step responses of system $T_1(s)$, system $T_2(s)$, and system $T_3(s)$

the pure second-order system response; $c_3(t)$, with a third pole close to the dominant poles, yields the most error.

4.7 System Response with Additional Poles

PROBLEM: Determine the validity of a second-order approximation for each of these two transfer functions:

a. $G(s) = \dfrac{700}{(s+15)(s^2+4s+100)}$

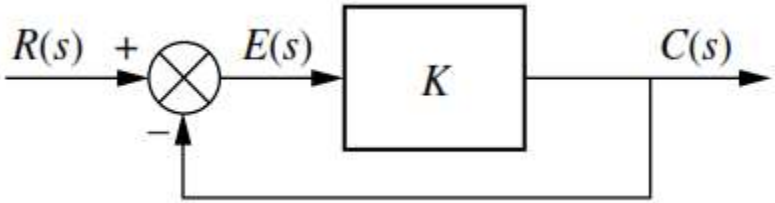
b. $G(s) = \dfrac{360}{(s+4)(s^2+2s+90)}$

ANSWERS:

a. The second-order approximation is valid.

b. The second-order approximation is not valid.

Lecture: Steady State Error



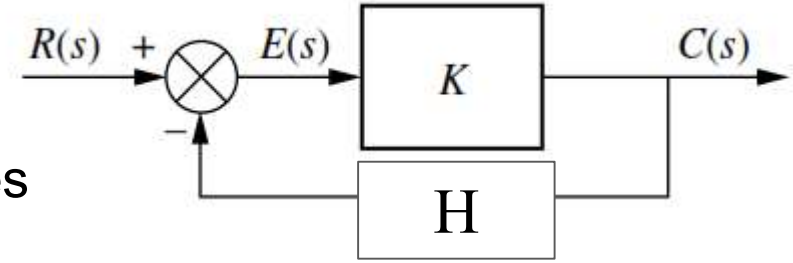
Steady State Error

Standard Test Inputs

Source of Errors

Steady State Error for Unity Feedback System

- Based on T(s)
- Based on G(s)



Static Error Constants and System Types

System Specifications

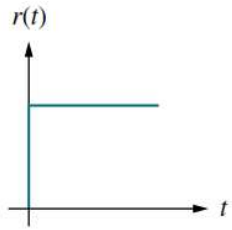
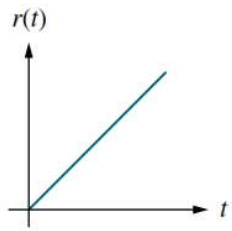
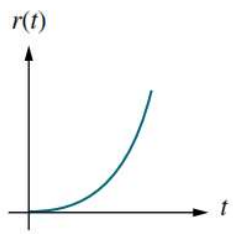
Steady State Error for disturbance

Steady State Error for NON-UNITY feedback systems

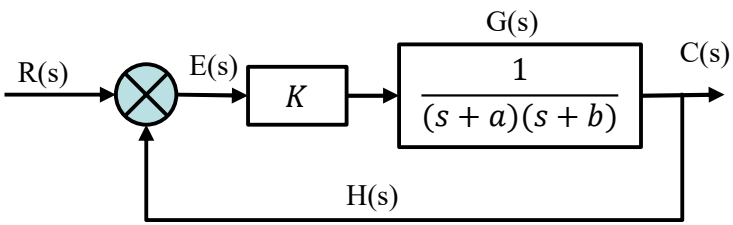
Lecture: Steady State Error

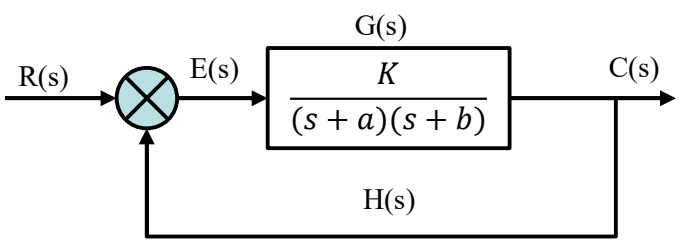
Steady-state error is the difference between the input and the output for a prescribed test input as $t \to \infty$. Test inputs used for steady-state error analysis and design are summarized in Table 7.1.

TABLE 7.1 Test waveforms for evaluating steady-state errors of position control systems

Waveform	Name	Physical interpretation	Time function	Laplace transform
	Step	Constant position	1	$\frac{1}{s}$
	Ramp	Constant velocity	t	$\frac{1}{s^2}$
	Parabola	Constant acceleration	$\frac{1}{2}t^2$	$\frac{1}{s^3}$

Lecture: Steady State Error





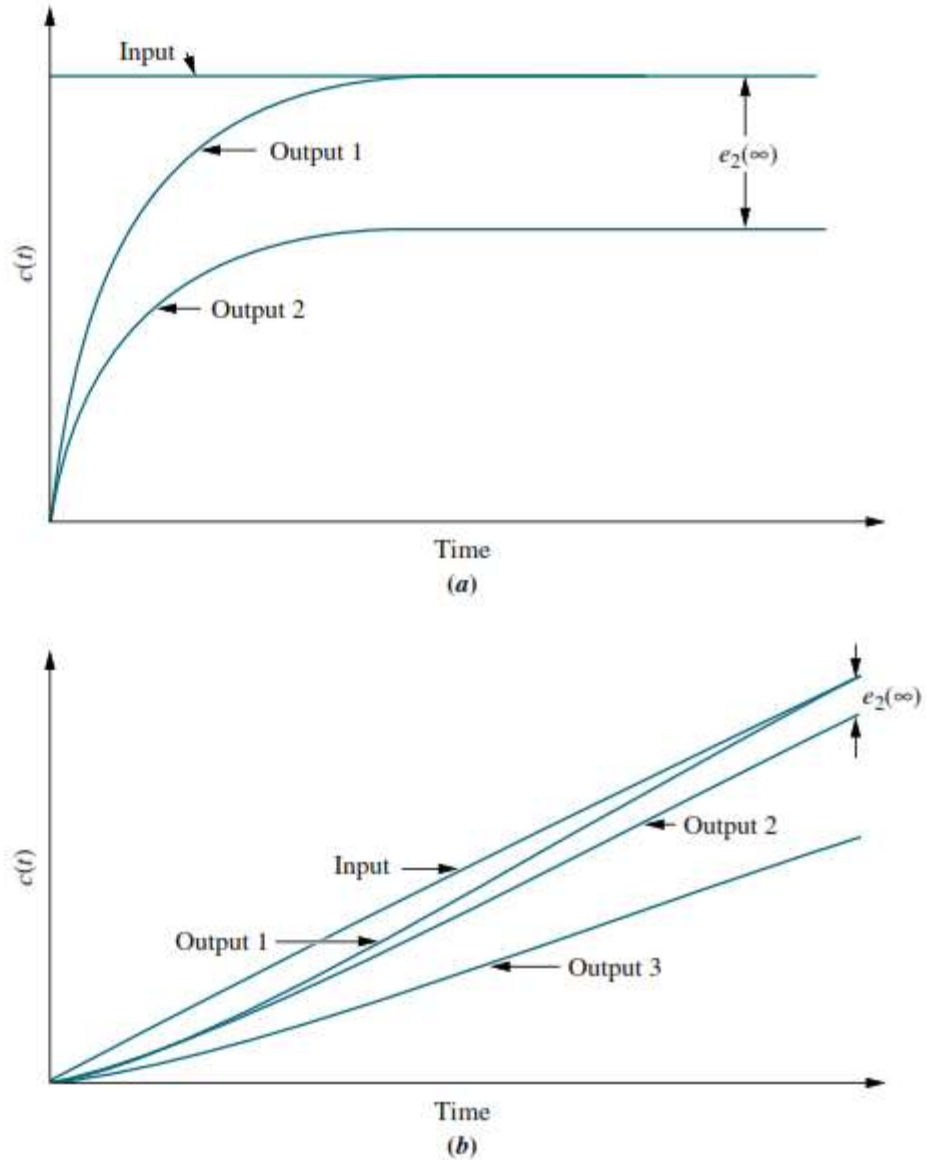
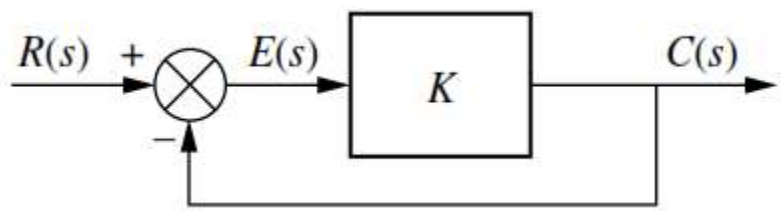


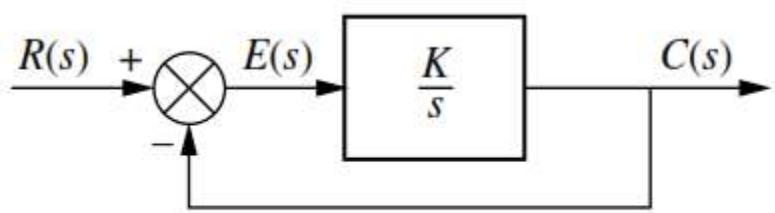
FIGURE 7.2 Steady-state error: **a.** step input; **b.** ramp input

Lecture: Sources of SSE

Non-linear sources
Backlash in gears
Motor>Threshold Voltage

$$e_{\text{steady-state}} = \frac{1}{K} c_{\text{steady-state}}$$



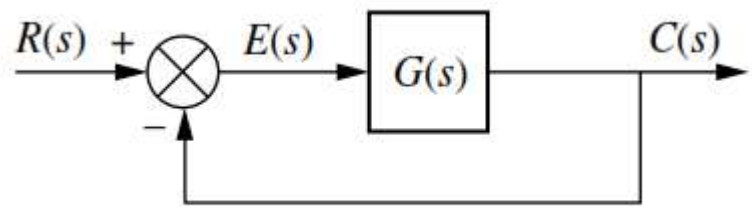


(b)

Lecture: SSE for Unity Feedback Systems

Based on:

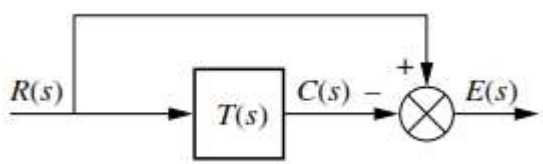
T(s), G(s)



(b)

$$E(s) = \frac{R(s)}{1 + G(s)}$$

$$\boxed{e(\infty) = \lim_{s \to 0} \frac{sR(s)}{1 + G(s)}}$$



$$E(s) = R(s)[1 - T(s)]$$

$$e(\infty) = \lim_{t \to \infty} e(t) = \lim_{s \to 0} sE(s)$$

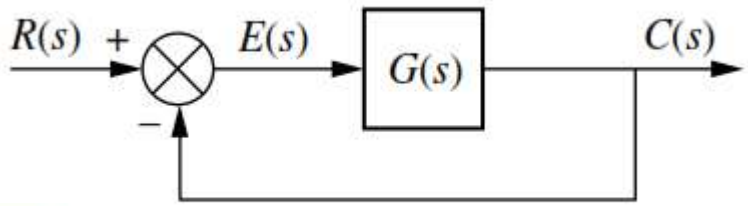
$$e(\infty) = \lim_{s \to \infty} sR(s)[1 - T(s)]$$

Valid only if (i) E(s) has LHS poles and poles at origin (ii) T(s) is stable.

Lecture: SSE for Unity Feedback Systems

Based on: G(s)

Integrator, n>=1, n>=2, n>=3



$$e(\infty) = e_{\text{step}}(\infty) = \lim_{s\to 0} \frac{s(1/s)}{1+G(s)} = \frac{1}{1+\lim_{s\to 0} G(s)}$$

$$e(\infty) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s\to 0} \frac{s(1/s^2)}{1+G(s)} = \lim_{s\to 0} \frac{1}{s+sG(s)} = \frac{1}{\lim_{s\to 0} sG(s)}$$

$$e(\infty) = e_{\text{parabola}}(\infty) = \lim_{s\to 0} \frac{s(1/s^3)}{1+G(s)} = \lim_{s\to 0} \frac{1}{s^2+s^2G(s)} = \frac{1}{\lim_{s\to 0} s^2G(s)}$$

Example 7.2

Steady-State Errors for Systems with No Integrations

PROBLEM: Find the steady-state errors for inputs of $5u(t)$, $5tu(t)$, and $5t^2u(t)$ to the system shown in Figure 7.5. The function $u(t)$ is the unit step.

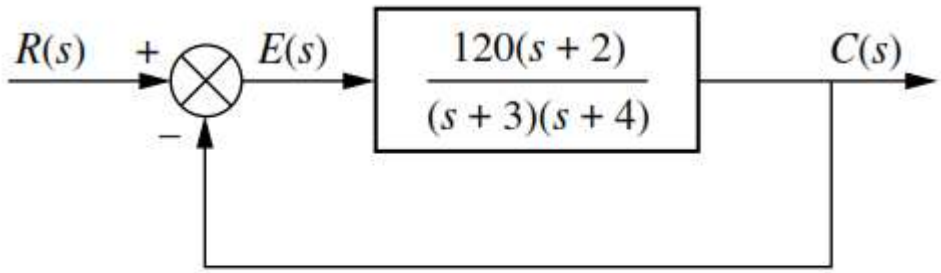


FIGURE 7.5 Feedback control system for Example 7.2

$$e(\infty) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$= \frac{1}{1+\lim_{s\to 0} G(s)}$$

$$= \frac{1}{\lim_{s\to 0} sG(s)}$$

$$= \frac{1}{\lim_{s\to 0} s^2G(s)}$$

$$e(\infty) = e_{\text{step}}(\infty) = \frac{5}{1+\lim_{s\to 0} G(s)} = \frac{5}{1+20} = \frac{5}{21}$$

$$e(\infty) = e_{\text{ramp}}(\infty) = \frac{5}{\lim_{s\to 0} sG(s)} = \frac{5}{0} = \infty$$

$$e(\infty) = e_{\text{parabola}}(\infty) = \frac{10}{\lim_{s\to 0} s^2G(s)} = \frac{10}{0} = \infty$$

Example 7.2

Steady-State Errors for Systems with No Integrations

PROBLEM: Find the steady-state errors for inputs of $5u(t)$, $5tu(t)$, and $5t^2u(t)$ to the system shown in Figure 7.5. The function $u(t)$ is the unit step.

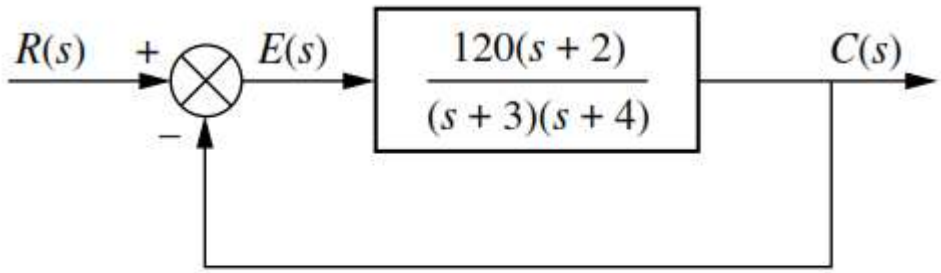


FIGURE 7.5 Feedback control system for Example 7.2

$$e(\infty) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$= \frac{1}{1+\lim_{s\to 0} G(s)}$$

$$= \frac{1}{\lim_{s\to 0} sG(s)}$$

$$= \frac{1}{\lim_{s\to 0} s^2G(s)}$$

$$e(\infty) = e_{\text{step}}(\infty) = \frac{5}{1+\lim_{s\to 0} G(s)} = \frac{5}{1+20} = \frac{5}{21}$$

$$e(\infty) = e_{\text{ramp}}(\infty) = \frac{5}{\lim_{s\to 0} sG(s)} = \frac{5}{0} = \infty$$

$$e(\infty) = e_{\text{parabola}}(\infty) = \frac{10}{\lim_{s\to 0} s^2G(s)} = \frac{10}{0} = \infty$$

Lecture: SSE for Unity Feedback Systems

PROBLEM: Find the steady-state errors for inputs of $5u(t)$, $5tu(t)$, and $5t^2u(t)$ to the system shown in Figure 7.6. The function $u(t)$ is the unit step.

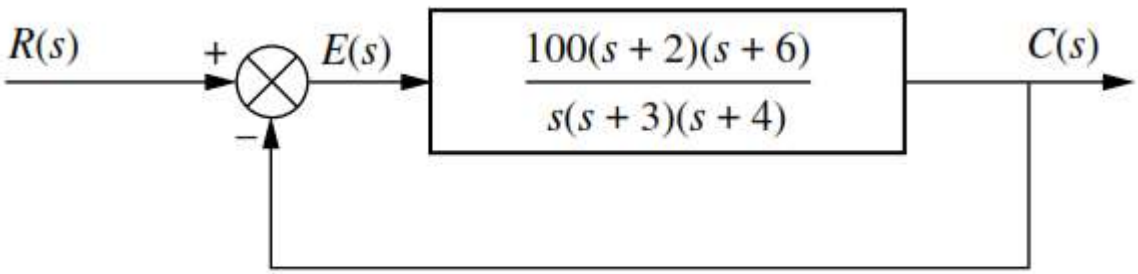


FIGURE 7.6 Feedback control system for Example 7.3

$$e(\infty) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

$$= \frac{1}{1+\lim_{s\to 0} G(s)}$$

$$= \frac{1}{\lim_{s\to 0} sG(s)}$$

$$= \frac{1}{\lim_{s\to 0} s^2G(s)}$$

$$e(\infty) = e_{\text{step}}(\infty) = \frac{5}{1+\lim_{s\to 0} G(s)} = \frac{5}{\infty} = 0$$

$$e(\infty) = e_{\text{ramp}}(\infty) = \frac{5}{\lim_{s\to 0} sG(s)} = \frac{5}{100} = \frac{1}{20}$$

$$e(\infty) = e_{\text{parabola}}(\infty) = \frac{10}{\lim_{s\to 0} s^2G(s)} = \frac{10}{0} = \infty$$

Lecture: Static Error Constant and System Types

The three terms in the denominator that are taken to the limit determine the steady-state error. We call these limits static error constants. Individually, their names are

$$e(\infty) = \lim_{s\to 0} \frac{sR(s)}{1+G(s)}$$

position constant, K_p, where

$$K_p = \lim_{s\to 0} G(s)$$

$$= \frac{1}{1+\lim_{s\to 0} G(s)}$$

velocity constant, K_v, where

$$K_v = \lim_{s\to 0} sG(s)$$

$$= \frac{1}{\lim_{s\to 0} sG(s)}$$

acceleration constant, K_a, where

$$K_a = \lim_{s\to 0} s^2G(s)$$

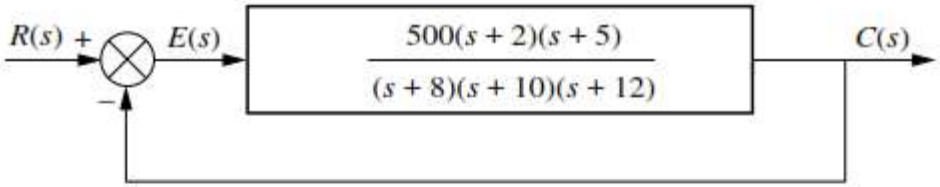
$$= \frac{1}{\lim_{s\to 0} s^2G(s)}$$

Steady-State Error via Static Error Constants

[redacted]

PROBLEM: For each system of Figure 7.7, evaluate the static error constants and find the expected error for the standard step, ramp, and parabolic inputs.

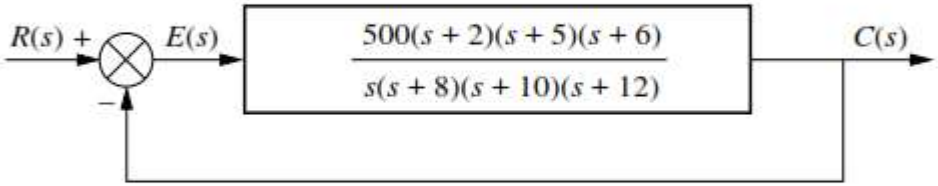
m Types



(a)

$$e(\infty) = \frac{1}{1 + K_p} = 0.161$$

$$e(\infty) = \lim_{s \to 0} \frac{sR(s)}{1 + G(s)}$$



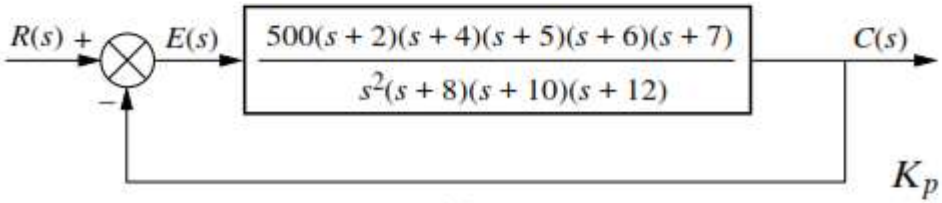
(b)

$$e(\infty) = \frac{1}{K_v} = \infty$$

$$= \frac{1}{1 + \lim_{s \to 0} G(s)}$$

$$e(\infty) = \frac{1}{K_a} = \infty$$

$$= \frac{1}{\lim_{s \to 0} sG(s)}$$



(c)

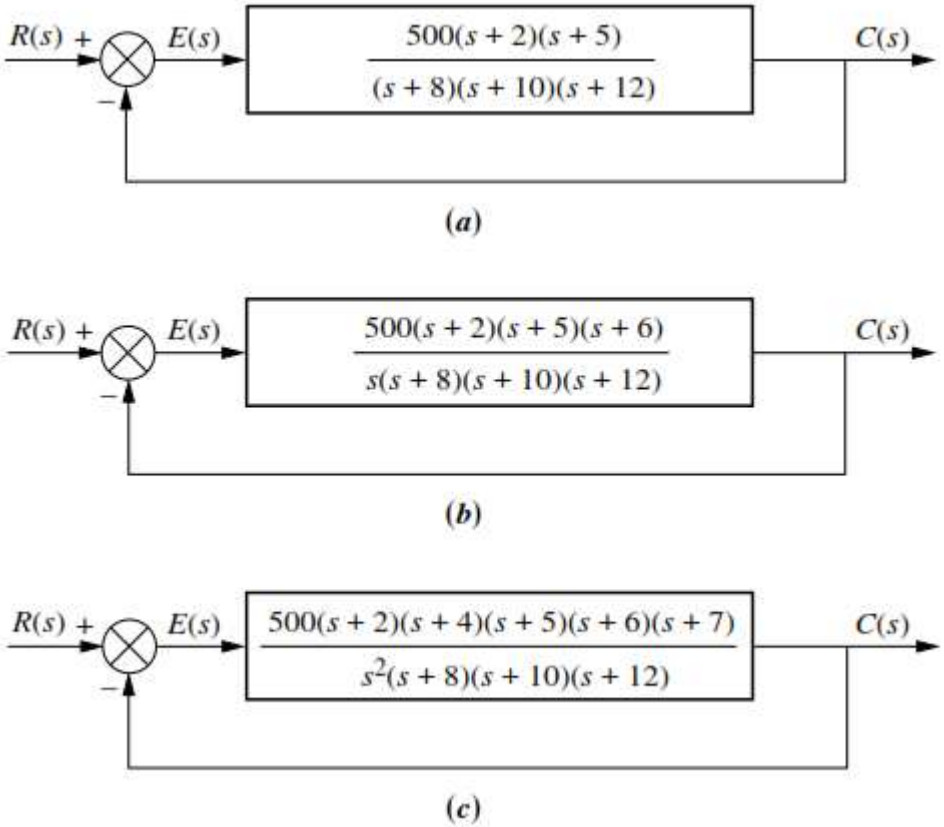
$$K_p = \lim_{s \to 0} G(s) = \frac{500 \times 2 \times 5}{8 \times 10 \times 12} = 5.208$$

$$= \frac{1}{\lim_{s \to 0} s^2 G(s)}$$

$$K_v = \lim_{s \to 0} sG(s) = 0$$

$$K_a = \lim_{s \to 0} s^2 G(s) = 0$$

Lecture: Static Error Constant and System Types



$$e(\infty) = \lim_{s \to 0} \frac{sR(s)}{1 + G(s)}$$

$$= \frac{1}{1 + \lim_{s \to 0} G(s)}$$

$$= \frac{1}{\lim_{s \to 0} sG(s)}$$

$$= \frac{1}{\lim_{s \to 0} s^2 G(s)}$$

Lecture: System Types

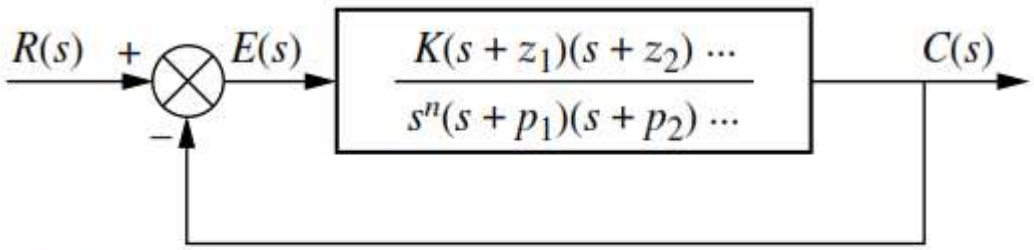


FIGURE 7.8 Feedback control system for defining system type

TABLE 7.2 Relationships between input, system type, static error constants, and steady-state errors

		Type 0		Type 1		Type 2	
Input	**Steady-state error formula**	**Static error constant**	**Error**	**Static error constant**	**Error**	**Static error constant**	**Error**
Step, $u(t)$	$\frac{1}{1+K_p}$	$K_p = \text{Constant}$	$\frac{1}{1+K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	∞	$K_v = \text{Constant}$	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = \text{Constant}$	$\frac{1}{K_a}$

Skill-Assessment Exercise 7.2

PROBLEM: A unity feedback system has the following forward transfer function:

$$G(s) = \frac{1000(s+8)}{(s+7)(s+9)}$$

a. Evaluate system type, K_p, K_v, and K_a.

b. Use your answers to **a.** to find the steady-state errors for the standard step, ramp, and parabolic inputs.

ANSWERS:

a. The closed-loop system is stable. System type = Type 0. $K_p = 127$, $K_v = 0$, and $K_a = 0$.

b. $e_{\text{step}}(\infty) = 7.8 \times 10^{-3}$, $e_{\text{ramp}}(\infty) = \infty$, and $e_{\text{parabola}}(\infty) = \infty$

Lecture: SSE Specifications

For example, if a control system has the specification $K_v = 1000$, we can draw several conclusions:

1. The system is stable.
2. The system is of Type 1, since only Type 1 systems have K_v's that are finite constants. Recall that $K_v = 0$ for Type 0 systems, whereas $K_v = \infty$ for Type 2 systems.
3. A ramp input is the test signal. Since K_v is specified as a finite constant, and the steady-state error for a ramp input is inversely proportional to K_v, we know the test input is a ramp.
4. The steady-state error between the input ramp and the output ramp is $1/K_v$ per unit of input slope.

Example 7.5

Interpreting the Steady-State Error Specification

PROBLEM: What information is contained in the specification $K_p = 1000$?

Example 7.6

Gain Design to Meet a Steady-State Error Specification

PROBLEM: Given the control system in Figure 7.10, find the value of K so that there is 10% error in the steady state.

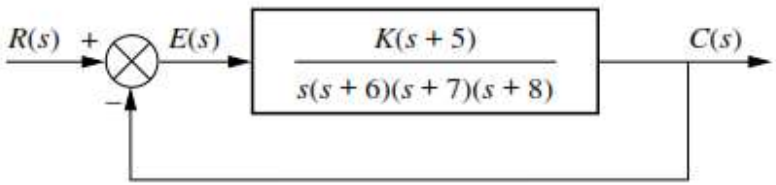


FIGURE 7.10 Feedback control system for Example 7.6

SOLUTION: Since the system is Type 1, the error stated in the problem must apply to a ramp input; only a ramp yields a finite error in a Type 1 system. Thus,

$$e(\infty) = \frac{1}{K_v} = 0.1 \tag{7.55}$$

Therefore,

$$K_v = 10 = \lim_{s \to 0} sG(s) = \frac{K \times 5}{6 \times 7 \times 8} \tag{7.56}$$

which yields

$$K = 672 \tag{7.57}$$

Applying the Routh-Hurwitz criterion, we see that the system is stable at this gain.

Although this gain meets the criteria for steady-state error and stability, it may not yield a desirable transient response. In Chapter 9 we will design feedback control systems to meet all three specifications.

Lecture: SSE Specifications

PROBLEM: A unity feedback system has the following forward transfer function:

$$G(s) = \frac{K(s+12)}{(s+14)(s+18)}$$

Find the value of K to yield a 10% error in the steady state.

ANSWER: $K = 189$

Lecture: SSE for disturbances

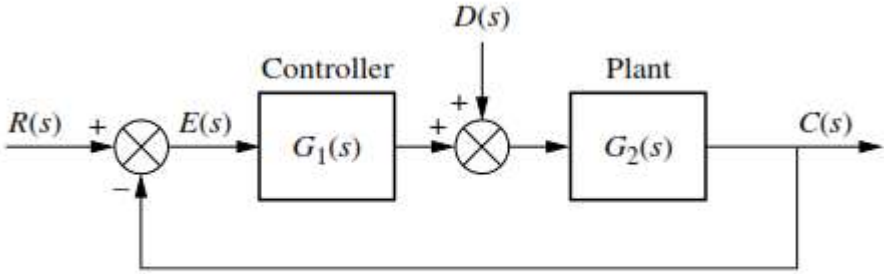


FIGURE 7.11 Feedback control system showing disturbance

$$C(s) = E(s)G_1(s)G_2(s) + D(s)G_2(s)$$

$$C(s) = R(s) - E(s)$$

$$E(s) = \frac{1}{1 + G_1(s)G_2(s)}R(s) - \frac{G_2(s)}{1 + G_1(s)G_2(s)}D(s)$$

where we can think of $1/[1 + G_1(s)G_2(s)]$ as a transfer function relating $E(s)$ to $R(s)$ and $-G_2(s)/[1 + G_1(s)G_2(s)]$ as a transfer function relating $E(s)$ to $D(s)$.

$$\begin{aligned} e(\infty) = \lim_{s\to 0} sE(s) &= \lim_{s\to 0} \frac{s}{1 + G_1(s)G_2(s)}R(s) - \lim_{s\to 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)}D(s) \\ &= e_R(\infty) + e_D(\infty) \end{aligned}$$

$$e_R(\infty) = \lim_{s\to 0} \frac{s}{1 + G_1(s)G_2(s)}R(s)$$

$$e_D(\infty) = -\lim_{s\to 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)}D(s)$$

Lecture: SSE for disturbances

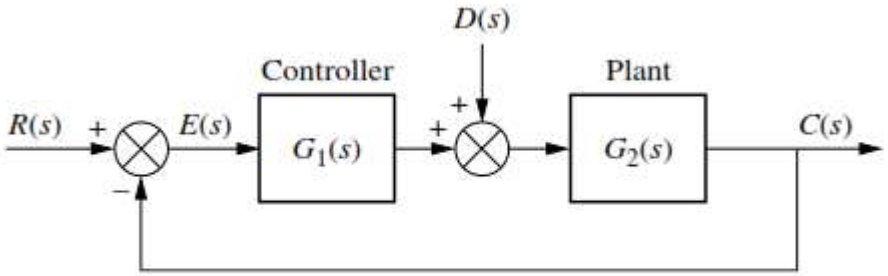


FIGURE 7.11 Feedback control system showing disturbance

$$e_D(\infty) = -\lim_{s\to 0} \frac{sG_2(s)}{1+G_1(s)G_2(s)} D(s)$$

$$e_D(\infty) = -\frac{1}{\lim_{s\to 0}\frac{1}{G_2(s)} + \lim_{s\to 0} G_1(s)}$$

For Step Input

This equation shows that the steady-state error produced by a step disturbance can be reduced by increasing the dc gain of $G_1(s)$ or decreasing the dc gain of $G_2(s)$.

Example 7.7

Steady-State Error Due to Step Disturbance

PROBLEM: Find the steady-state error component due to a step disturbance for the system of Figure 7.13.

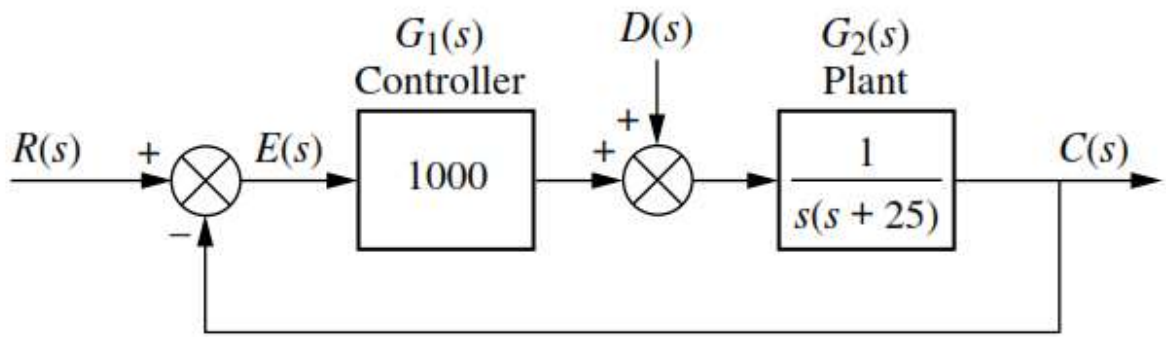
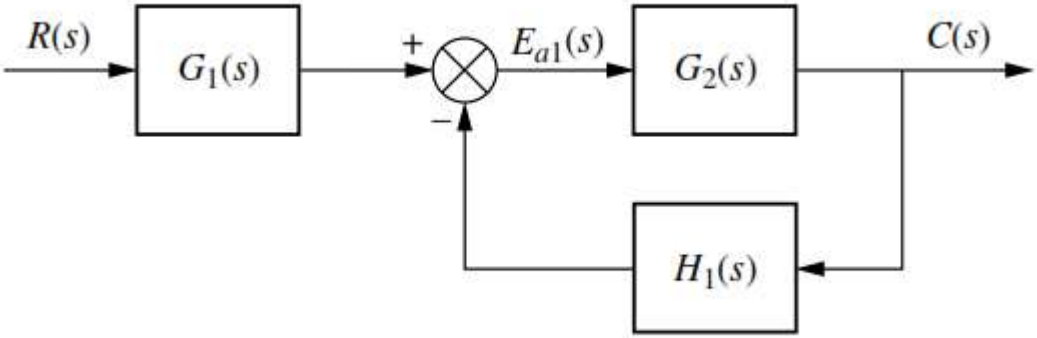


FIGURE 7.13 Feedback control system for Example 7.7

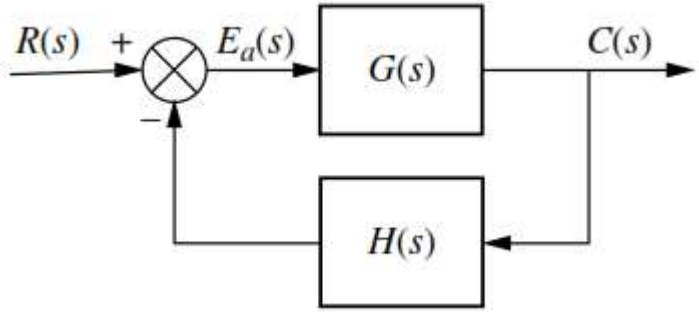
SOLUTION: The system is stable. Using Figure 7.12 and Eq. (7.62), we find

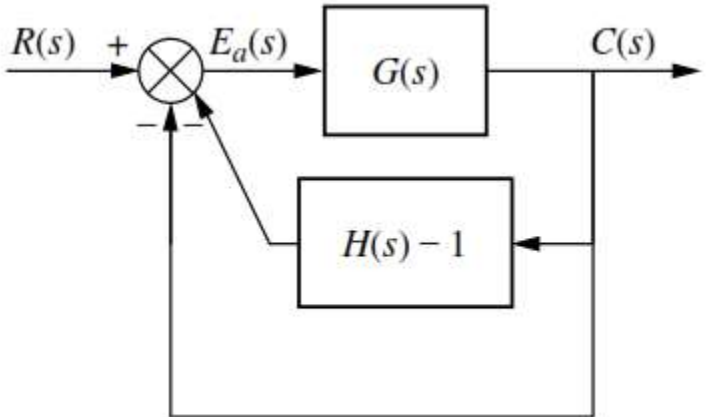
$$e_D(\infty) = -\frac{1}{\lim_{s\to 0}\frac{1}{G_2(s)} + \lim_{s\to 0} G_1(s)} = -\frac{1}{0+1000} = -\frac{1}{1000} \tag{7.63}$$

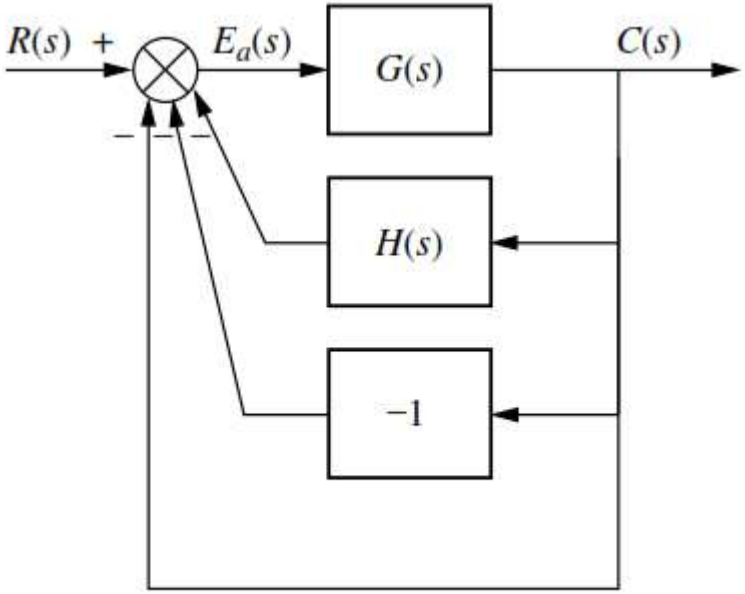
Lecture: SSE for Non-Unity Feedback Systems



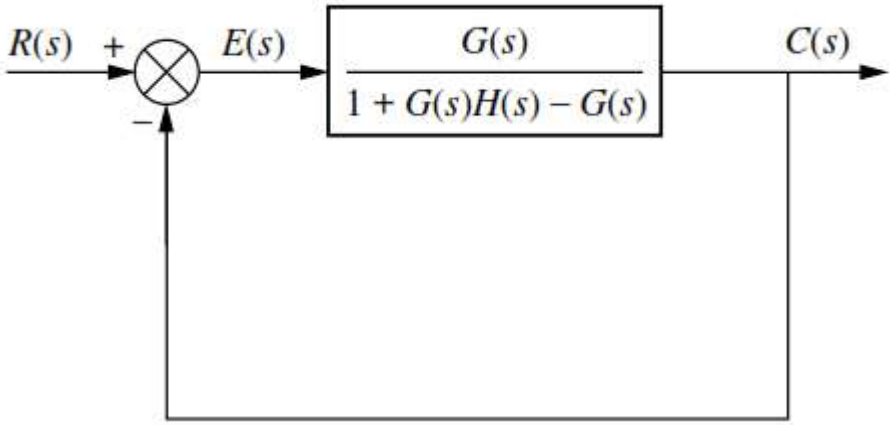
(a)



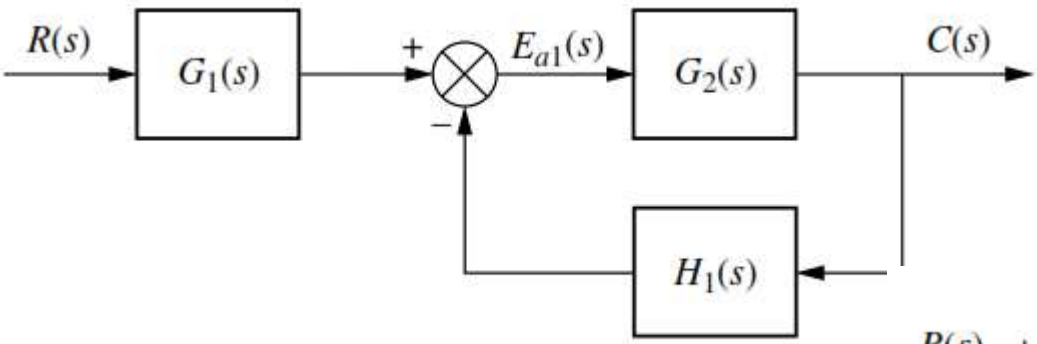




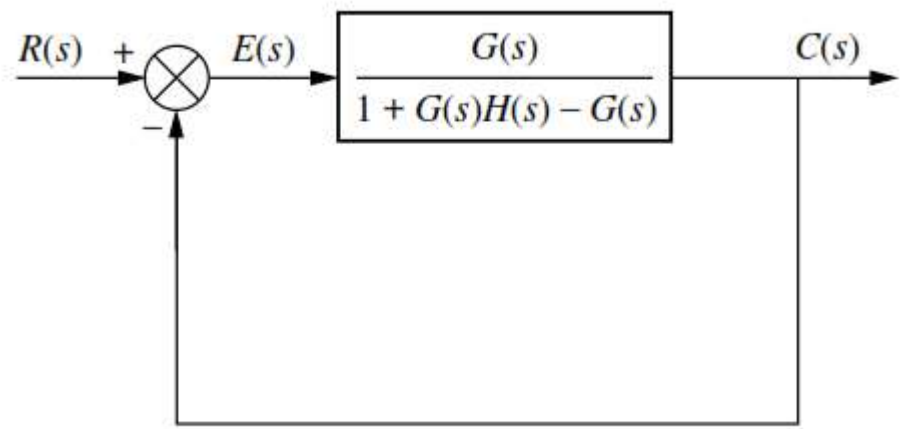
(c)



Lecture: SSE for Non-Unity Feedback Systems



(a)



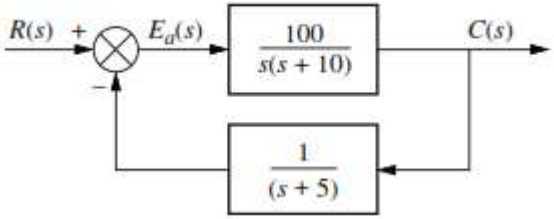
(e)

Lecture: SSE for Non-Unity Feedback Systems

Example 7.8

Steady-State Error for Nonunity Feedback Systems

PROBLEM: For the system shown in Figure 7.16, find the system type, the appropriate error constant associated with the system type, and the steady-state error for a unit step input. Assume input and output units are the same.



$$G_e(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)} = \frac{100(s+5)}{s^3 + 15s^2 - 50s - 400}$$

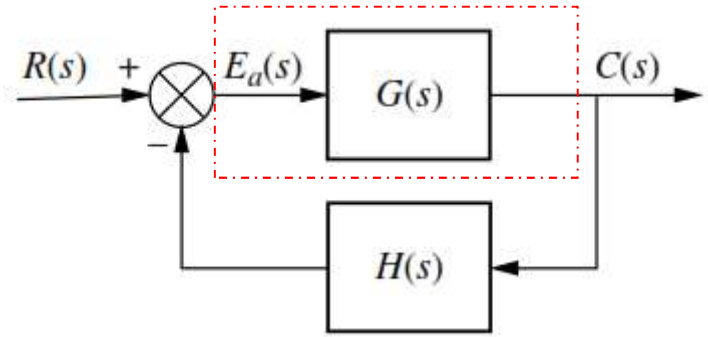
$$K_p = \lim_{s \to 0} G_e(s) = \frac{100 \times 5}{-400} = -\frac{5}{4}$$

$$e(\infty) = \frac{1}{1 + K_p} = \frac{1}{1 - (5/4)} = -4$$

Lecture: Root Locus

G(s)- Open Loop TF
H(s)- Feedback TF
T(s)- Closed Loop TF
K- Gain

Forward TF



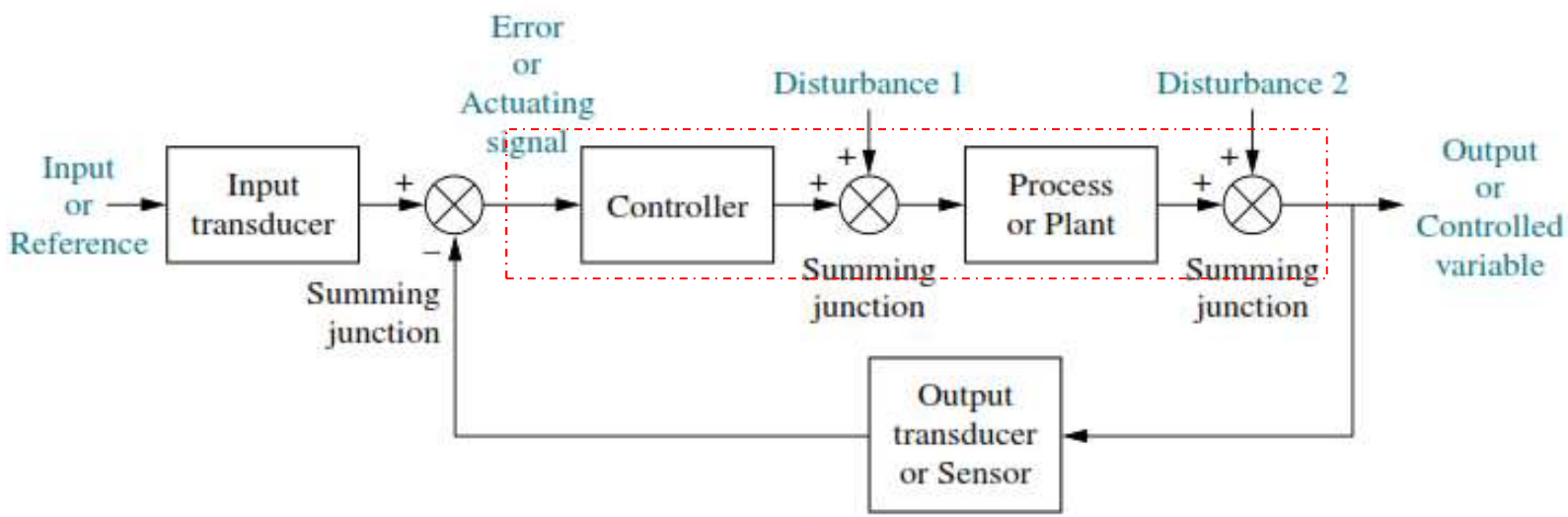
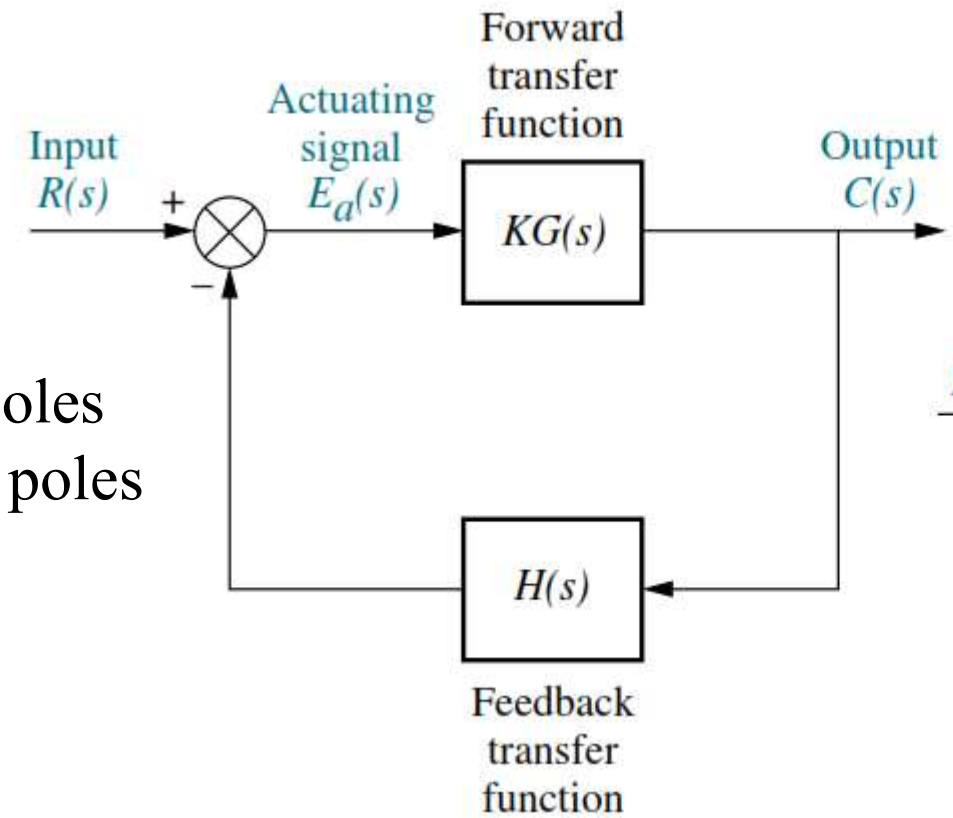
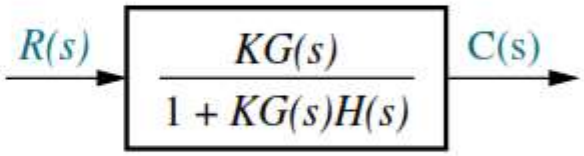


FIGURE 1.6 Block diagrams of control systems: **a.** open-loop system; **b.** closed-loop system

Lecture: Root Locus

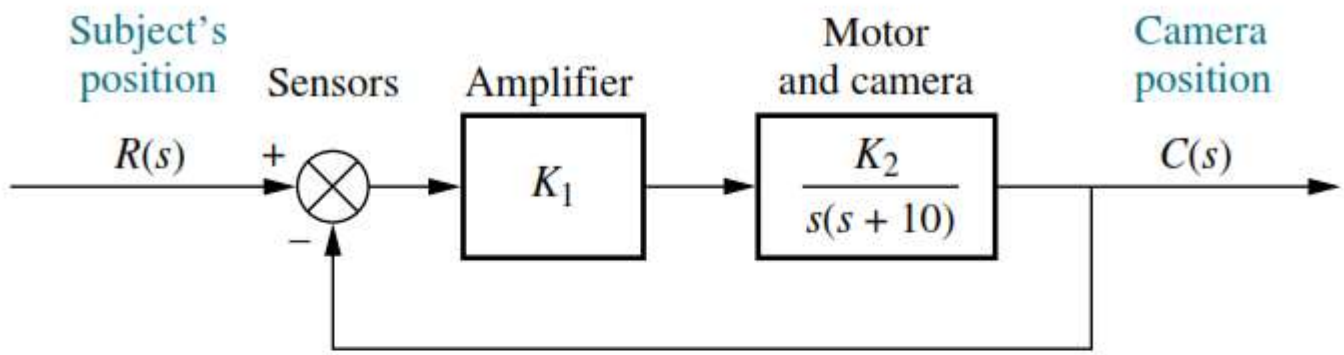




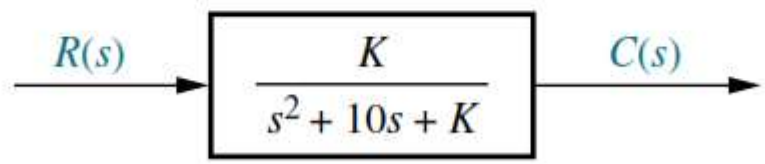
Open loop poles
Closed loop poles

$$T(s) = \frac{KN_G(s)D_H(s)}{D_G(s)D_H(s) + KN_G(s)N_H(s)}$$

Lecture: Root Locus



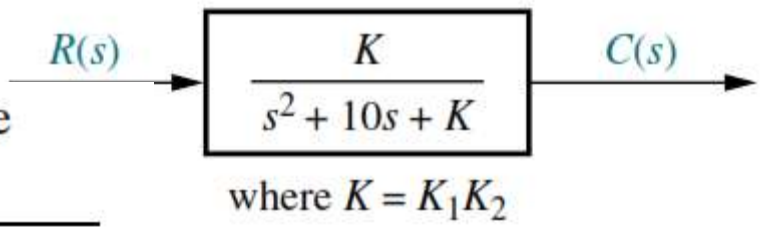
(b)



where $K = K_1K_2$

(c)

Lecture: Root Locus



(c)

TABLE 8.1 Pole location as function of gain for the system of Figure 8.4

K	Pole 1	Pole 2
0	-10	0
5	-9.47	-0.53
10	-8.87	-1.13
15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5 + j2.24$	$-5 - j2.24$
35	$-5 + j3.16$	$-5 - j3.16$
40	$-5 + j3.87$	$-5 - j3.87$
45	$-5 + j4.47$	$-5 - j4.47$
50	$-5 + j5$	$-5 - j5$

Lecture: Root Locus

TABLE 8.1 Pole location as function of gain for the system of Figure 8.4

K	Pole 1	Pole 2
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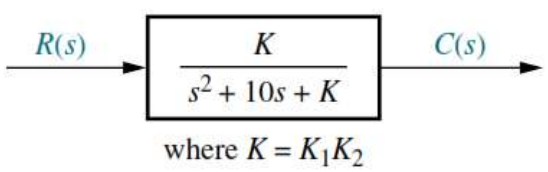
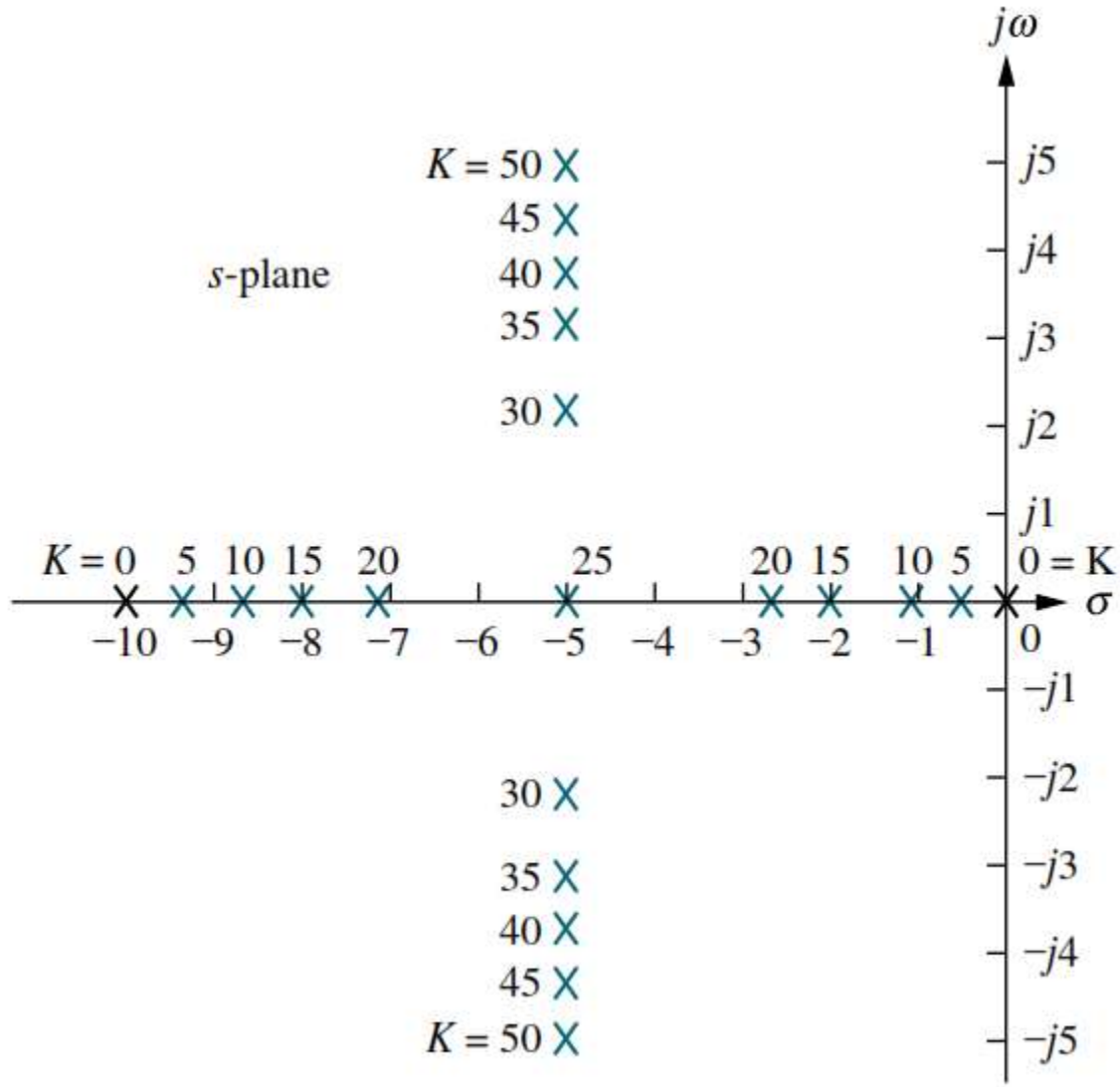
30	$-5+j2.24$	$-5-j2.24$
35	$-5+j3.16$	$-5-j3.16$
40	$-5+j3.87$	$-5-j3.87$
45	$-5+j4.47$	$-5-j4.47$
50	$-5+j5$	$-5-j5$



where $K = K_1K_2$

(c)



(a)

Lecture: Root Locus

TABLE 8.1 Pole location as function of gain for the system of Figure 8.4

K	Pole 1	Pole 2
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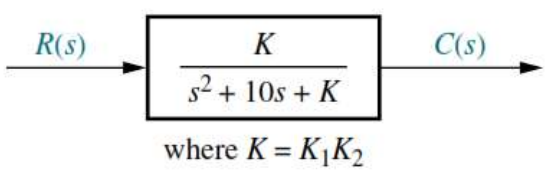
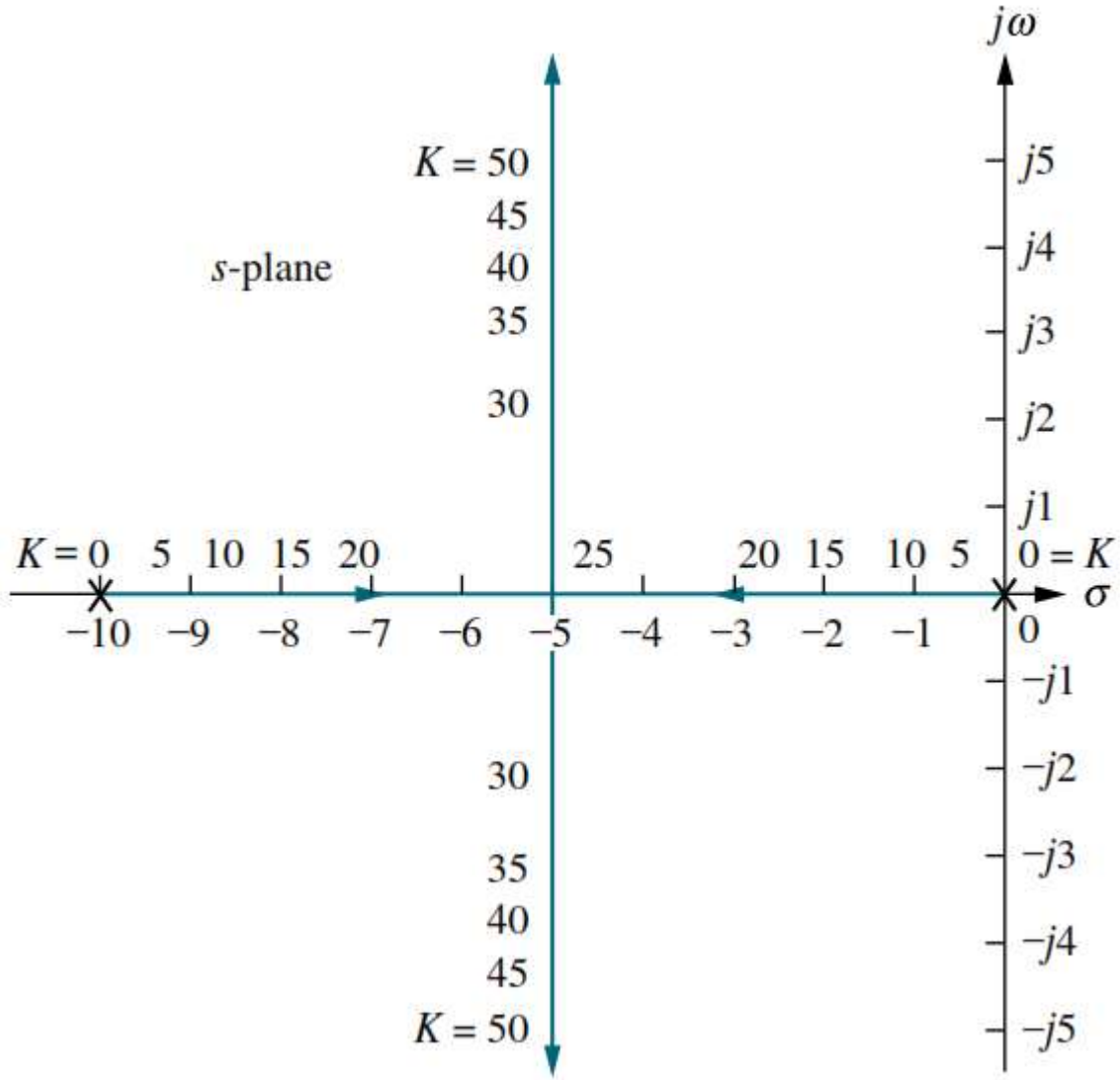
50	$-5+j5$	$-5-j5$



where $K = K_1K_2$

(c)



(b)

Lecture: Root Locus

TABLE 8.1 Pole location as function of gain for the system of Figure 8.4

K	Pole 1	Pole 2
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15	-8.16	-1.84
20	-7.24	-2.76
25	-5	-5
30	$-5+j2.24$	$-5-j2.24$
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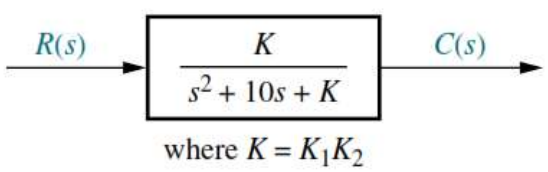
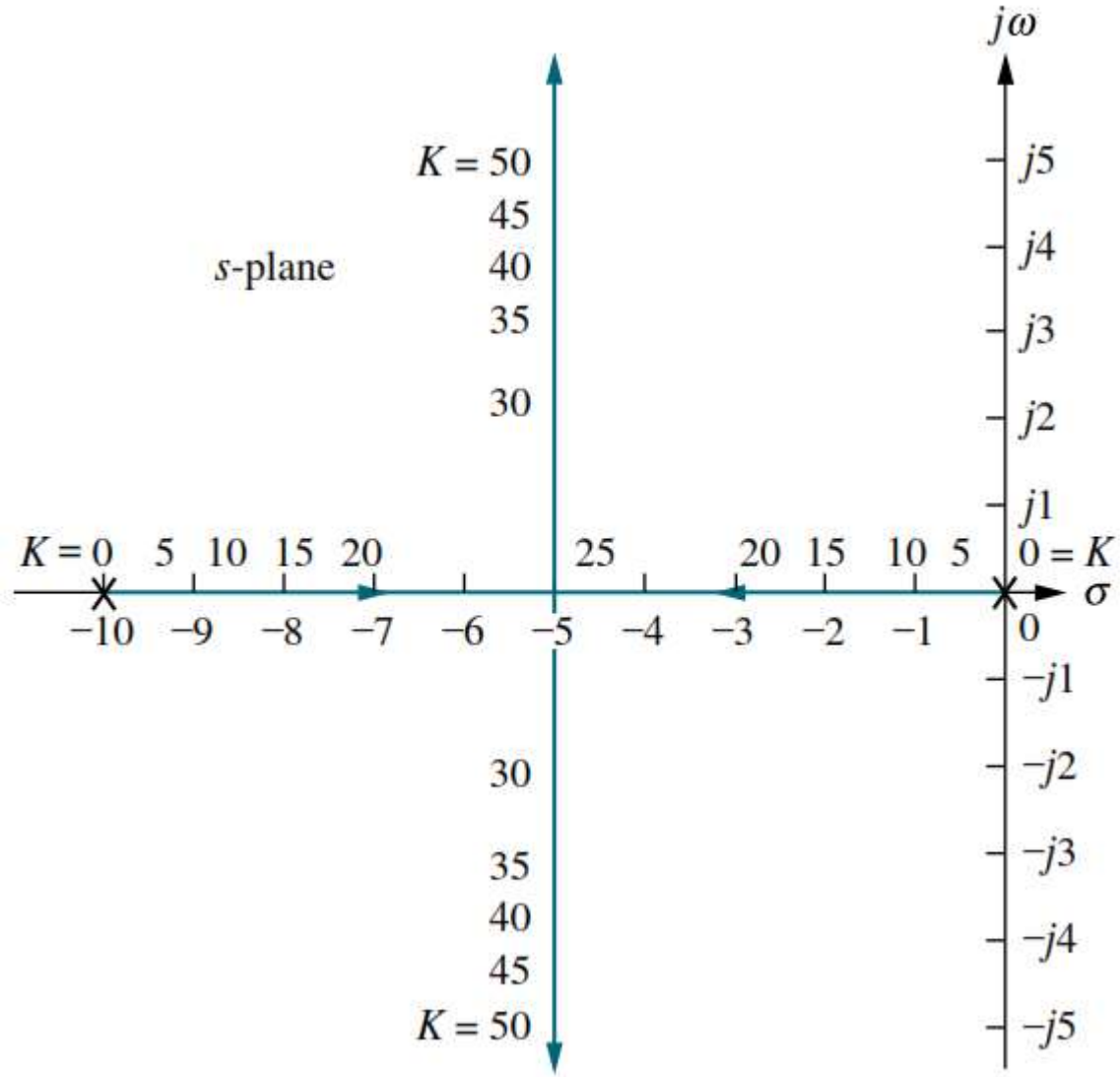
40	$-5+j3.87$	$-5-j3.87$
45	$-5+j4.47$	$-5-j4.47$
50	$-5+j5$	$-5-j5$



where $K = K_1K_2$

(c)



(b)

Lecture: Root Locus

Lecture: Root Locus

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$$

$$KG(s)H(s) = -1 = 1\angle(2k+1)180^\circ \quad k = 0, \pm 1, \pm 2, \pm 3, \ldots$$

$$|KG(s)H(s)| = 1$$

$$\angle KG(s)H(s) = (2k+1)180^\circ$$

$$K = \frac{1}{|G(s)||H(s)|}$$

Lecture: Root Locus

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$$

$$|KG(s)H(s)| = 1$$

$$\angle KG(s)H(s) = (2k+1)180^\circ$$

$$K = \frac{1}{|G(s)||H(s)|}$$

$$F(s) = \frac{\prod_{i=1}^{m}(s+z_i)}{\prod_{j=1}^{n}(s+p_j)} = \frac{\prod \text{numerator's complex factors}}{\prod \text{denominator's complex factors}}$$

$$M = \frac{\prod \text{zero lengths}}{\prod \text{pole lengths}} = \frac{\prod_{i=1}^{m}|(s+z_i)|}{\prod_{j=1}^{n}|(s+p_j)|}$$

$$\begin{aligned}\theta &= \sum_{m} \text{zero angles} - \sum_{n} \text{pole angles} \\ &= \sum_{i=1}^{m} \angle(s+z_i) - \sum_{j=1}^{n} \angle(s+p_j)\end{aligned}$$

Lecture: Root Locus

$$T(s) = \frac{KG(s)}{1 + KG(s)H(s)}$$

$$|KG(s)H(s)| = 1$$

$$\angle KG(s)H(s) = (2k+1)180^\circ$$

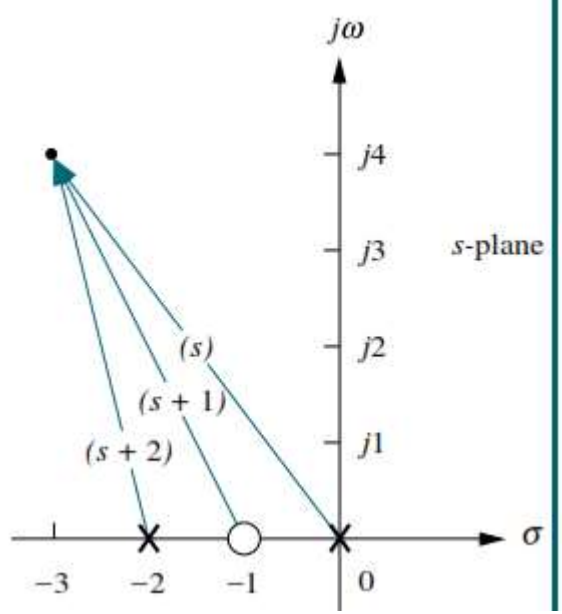
$$K = \frac{1}{|G(s)||H(s)|}$$

PROBLEM: Given

$$F(s) = \frac{(s+1)}{s(s+2)}$$

$$F(s) = \frac{\prod_{i=1}^{m}(s+z_i)}{\prod_{j=1}^{n}(s+p_j)} = \frac{\prod \text{numerator's complex factors}}{\prod \text{denominator's complex factors}}$$

find $F(s)$ at the point $s = -3 + j4$.



$$M = \frac{\prod \text{zero lengths}}{\prod \text{pole lengths}} = \frac{\prod_{i=1}^{m}|(s+z_i)|}{\prod_{j=1}^{n}|(s+p_j)|}$$

$$\theta = \sum_{m} \text{zero angles} - \sum_{n} \text{pole angles}$$
$$= \sum_{i=1}^{m} \angle(s+z_i) - \sum_{j=1}^{n} \angle(s+p_j)$$

Lecture: Root Locus

PROBLEM: Given a unity feedback system that has the forward transfer function

$$G(s) = \frac{K(s+2)}{(s^2+4s+13)}$$

do the following:

a. Calculate the angle of $G(s)$ at the point $(-3+j0)$ by finding the algebraic sum of angles of the vectors drawn from the zeros and poles of $G(s)$ to the given point.

b. Determine if the point specified in **a** is on the root locus.

c. If the point specified in **a** is on the root locus, find the gain, K, using the lengths of the vectors.

a. Sum of angles $= 180°$

b. Point is on the root locus

c. $K = 10$

Lecture: Root Locus

Sketching the Root Locus

Number of branches The number of branches of the root locus equals the number of closed-loop poles.

Symmetry The root locus is symmetrical about the real axis.

Real-axis segments On the real axis, for $K > 0$ the root locus exists to the left of an odd number of real-axis, finite open-loop poles and/or finite open-loop zeros.

Starting and ending points The root locus begins at the finite and infinite poles of $G(s)H(s)$ and ends at the finite and infinite zeros of $G(s)H(s)$.

Behavior at infinity The root locus approaches straight lines as asymptotes as the locus approaches infinity. Further, the equations of the asymptotes are given by the real-axis intercept and angle in radians as follows:

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\#\text{finite poles} - \#\text{finite zeros}}$$

$$\theta_a = \frac{(2k+1)\pi}{\#\text{finite poles} - \#\text{finite zeros}}$$

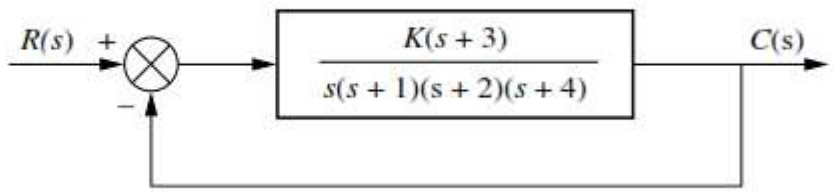
where $k = 0, \pm 1, \ \pm 2, \ \pm 3, \ldots$.

Lecture: Root Locus

Sketching the Root Locus

PROBLEM: Sketch the root locus for the system shown in Figure 8.11.

$(2k+1)\frac{\pi}{3}$



$-K$ $\quad 3$

-1.63

FIGURE 8.11 System for Example 8.2.

$k=0, \quad \theta_0 = \frac{\pi}{3} = 60^\circ$

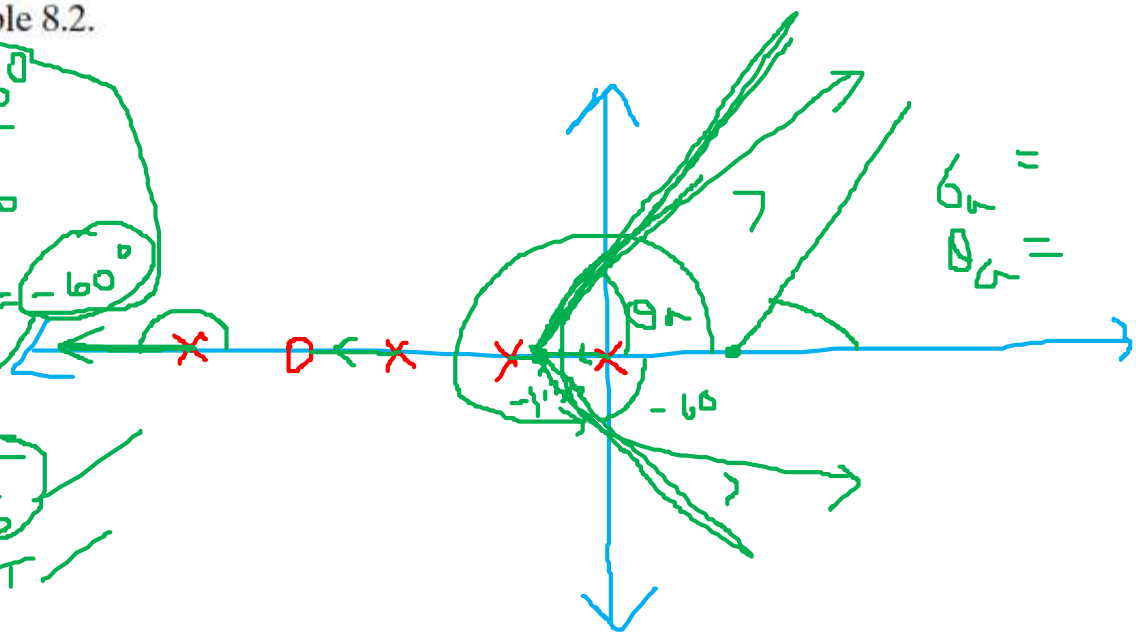
$k=1, \quad \theta_1 = \pi = 180^\circ$

$k=2, \quad \theta_2 = \frac{5\pi}{3} = 300^\circ = -60^\circ$

$k=3, \quad \theta_3 = \frac{7\pi}{3} = \frac{\pi}{3}$

$k=4, \quad = \frac{9\pi}{3} = \pi$

$k=5, \quad = \frac{11\pi}{3} = -60^\circ$



Lecture: Root Locus

Sketching the Root Locus

PROBLEM: Sketch the root locus for the system shown in Figure 8.11.

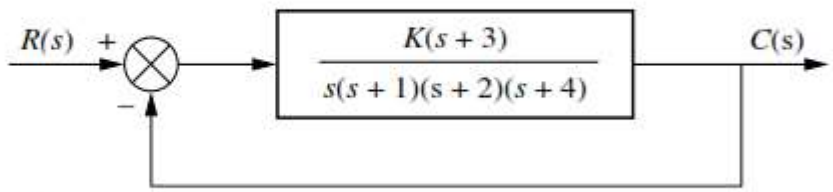


FIGURE 8.11 System for Example 8.2.

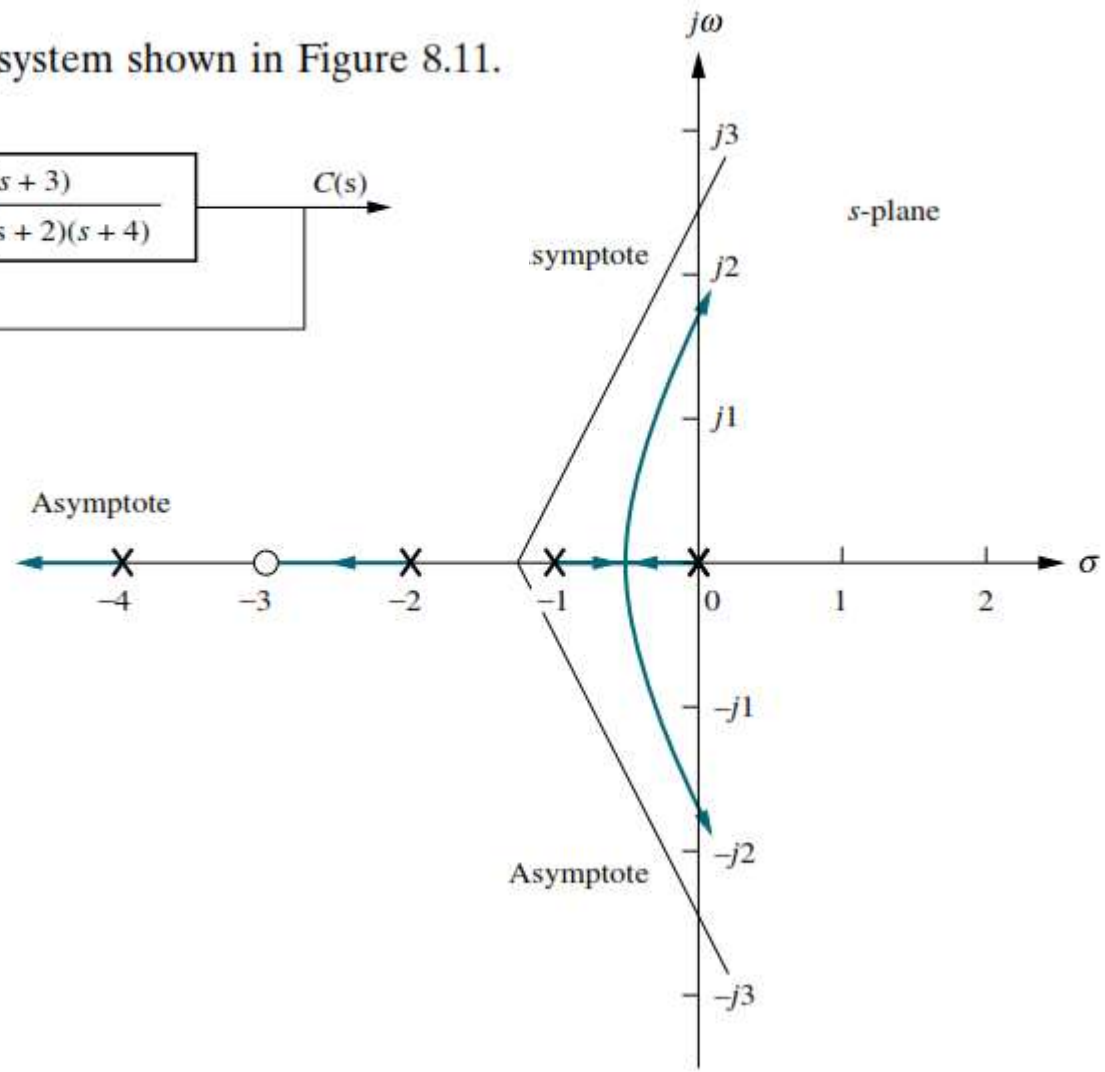
$$\sigma_a = \frac{(-1-2-4)-(-3)}{4-1} = -\frac{4}{3}$$

$$\theta_a = \frac{(2k+1)\pi}{\#\text{finite poles} - \#\text{finite zeros}}$$

$$= \pi/3 \qquad \text{for } k = 0$$

$$= \pi \qquad \text{for } k = 1$$

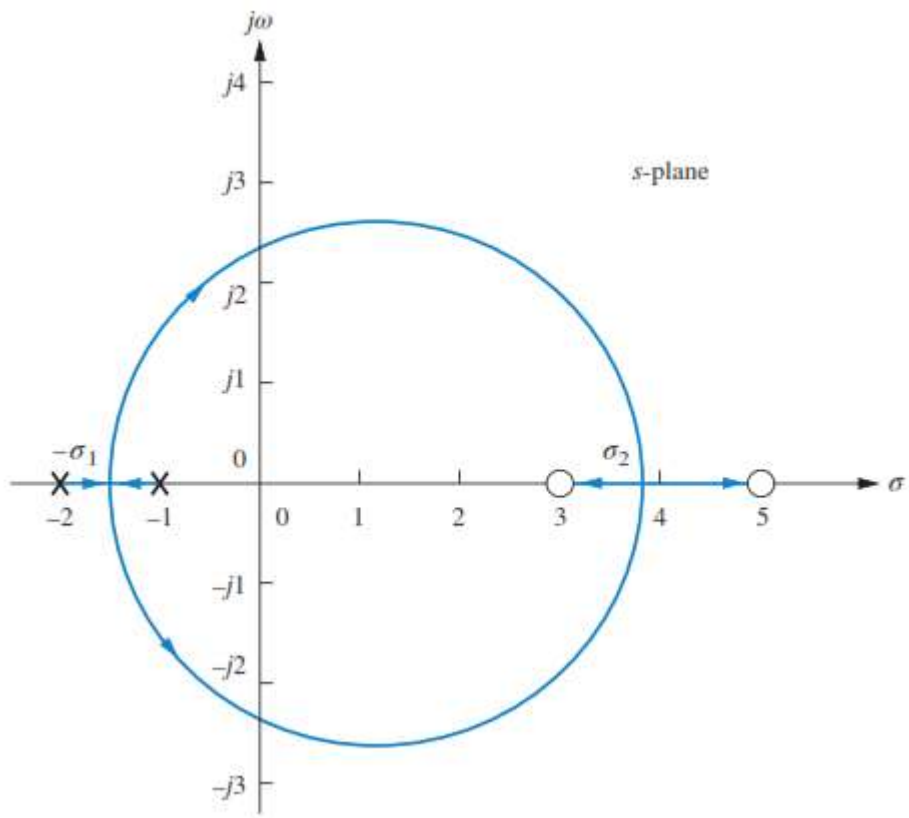
$$= 5\pi/3 \qquad \text{for } k = 2$$



Lecture: Root Locus

8.5 Refining the Sketch

Breakaway and Break-in Points
The jw axis crossing
Angles of departure and arrival



Lecture: Root Locus

8.5 Refining the Sketch

Breakaway and Break-in Points via Differentiation

PROBLEM: Find the breakaway and break-in points for the root locus of Figure 8.13, using differential calculus.

$$KG(s)H(s) = \frac{K(s-3)(s-5)}{(s+1)(s+2)} = \frac{K(s^2-8s+15)}{(s^2+3s+2)}$$

But for all points along the root locus, $KG(s)H(s) = -1$, and along the real axis, $s = \sigma$.

$$\frac{K(\sigma^2-8\sigma+15)}{(\sigma^2+3\sigma+2)} = -1 \qquad K = \frac{-(\sigma^2+3\sigma+2)}{(\sigma^2-8\sigma+15)}$$

Differentiating K with respect to σ and setting the derivative equal to zero yields

$$\frac{dK}{d\sigma} = \frac{(11\sigma^2-26\sigma-61)}{(\sigma^2-8\sigma+15)^2} = 0$$

Solving for σ, we find $\sigma = -1.45$ and 3.82, which are the breakaway and break-in points.

Lecture: Root Locus

8.5 Refining the Sketch

Breakaway and Break-in Points Without Differentiation

Transition Method

Breakaway and break-in points satisfy the relationship

$$\frac{1}{\sigma-3}+\frac{1}{\sigma-5}=\frac{1}{\sigma+1}+\frac{1}{\sigma+2}$$

$$\boxed{\sum_{1}^{m}\frac{1}{\sigma+z_i}=\sum_{1}^{n}\frac{1}{\sigma+p_i}} \tag{8.37}$$

where z_i and p_i are the negative of the zero and pole values, respectively, of $G(s)H(s)$.

$$\frac{1}{\sigma-3}+\frac{1}{\sigma-5}=\frac{1}{\sigma+1}+\frac{1}{\sigma+2}$$

$$11\sigma^2-26\sigma-61=0$$

Hence, $\sigma=-1.45$ and 3.82, which agrees with Example 8.3.

Lecture: Root Locus

8.5 Refining the Sketch

The $j\omega$-Axis Crossings

PROBLEM: For the system of Figure 8.11, find the frequency and gain, K, for which the root locus crosses the imaginary axis. For what range of K is the system stable?

The closed-loop transfer function for the system of Figure 8.11 is

$$T(s) = \frac{K(s+3)}{s^4 + 7s^3 + 14s^2 + (8+K)s + 3K}$$

A complete row of zeros yields the possibility for imaginary axis roots. For positive values of gain, those for which the root locus is plotted, only the s^1 row can yield a row of zeros. Thus,

$$-K^2 - 65K + 720 = 0$$

$$K = 9.65 > 0$$

Forming the even polynomial by using the s^2 row with $K = 9.65$,

$$(90 - K)s^2 + 21K = 80.35s^2 + 202.7 = 0$$

and s is found to be equal to $\pm j1.59$.

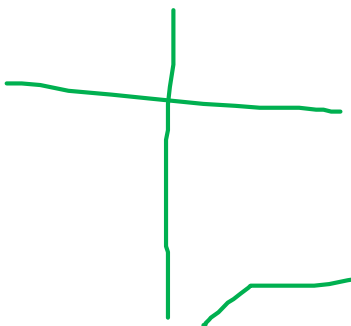
TABLE 8.3 Routh table for Eq (8.40)

s^4	1	14	$3K$
s^3	7	$8+K$	
s^2	$90-K$	$21K$	
s^1	$\dfrac{-K^2-65K+720}{90-K}$		
s^0	$21K$		

Lecture: Root Locus

8.5 Refining the Sketch

Angles of Departure and Arrival



K_G

1. Root locus
2. K
3. $j\omega$
4. $\leftrightarrow$

$G(s) = \frac{s}{s+3}$ $\frac{Y\angle\theta}{Y\angle\theta}$

$s = -3 + 5j$

$|G| = ?$

$\angle G = ?$

$\text{odd} \times 180°$

Lecture: Design via Root Locus

$$G_T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

K

$$s_{1,2} = -\sigma_d \pm j\omega_d$$

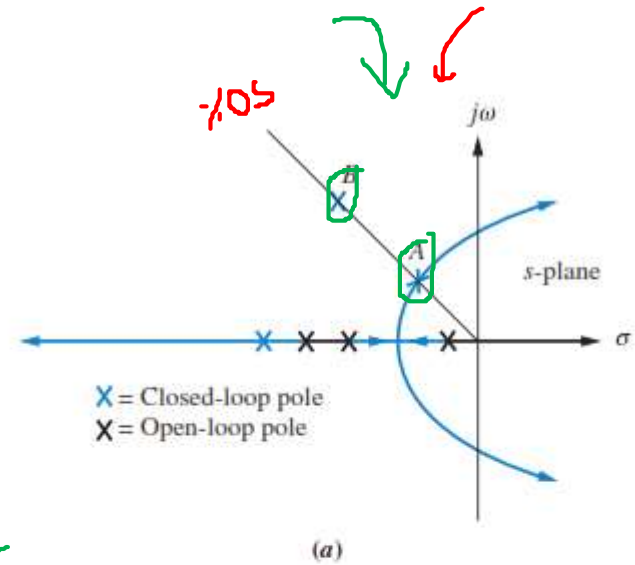
$$\sigma_d = \zeta\omega_n$$

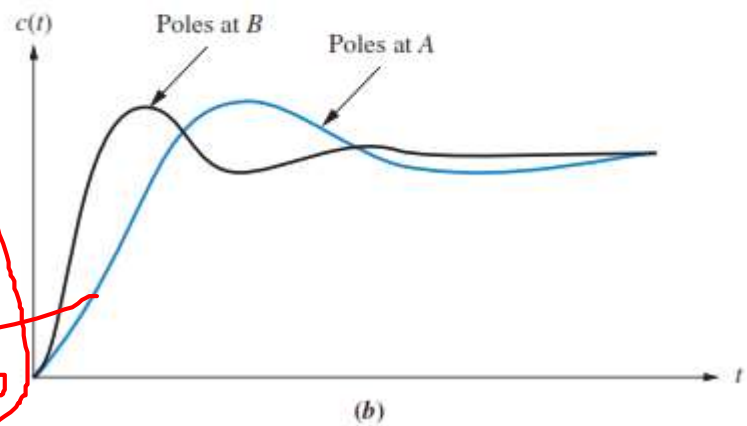
$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\rightarrow T_s = \frac{4}{\sigma_d}$$

$$\rightarrow T_p = \frac{\pi}{\omega_d}$$

1.0s





$T_p = 1.0\,s \approx 5\,s$

$T_s = 2.0\,s \approx 8\,s$

Lecture: Design via Root Locus

TABLE 9.7 Types of cascade compensators

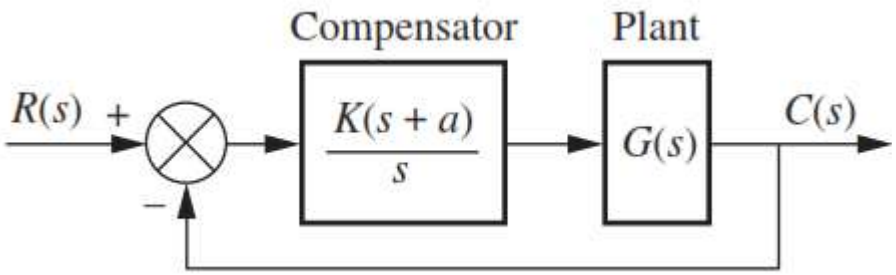
Function	Compensator	Transfer function	Characteristics
Improve steady-state error	PI	$K\frac{s+z_c}{s}$	**1.** Increases system type. **2.** Error becomes zero. **3.** Zero at $-z_c$ is small and negative. **4.** Active circuits are required to implement.
Improve steady-state error	Lag	$K\frac{s+z_c}{s+p_c}$	**1.** Error is improved but not driven to zero. **2.** Pole at $-p_c$ is small and negative. **3.** Zero at $-z_c$ is close to, and to the left of, the pole at $-p_c$. **4.** Active circuits are not required to implement.
Improve transient response	PD	$K(s+z_c)$	**1.** Zero at $-z_c$ is selected to put design point on root locus. **2.** Active circuits are required to implement. **3.** Can cause noise and saturation; implement with rate feedback or with a pole (lead).
Improve transient response	Lead	$K\frac{s+z_c}{s+p_c}$	**1.** Zero at $-z_c$ and pole at $-p_c$ are selected to put design point on root locus. **2.** Pole at $-p_c$ is more negative than zero at $-z_c$. **3.** Active circuits are not required to implement.

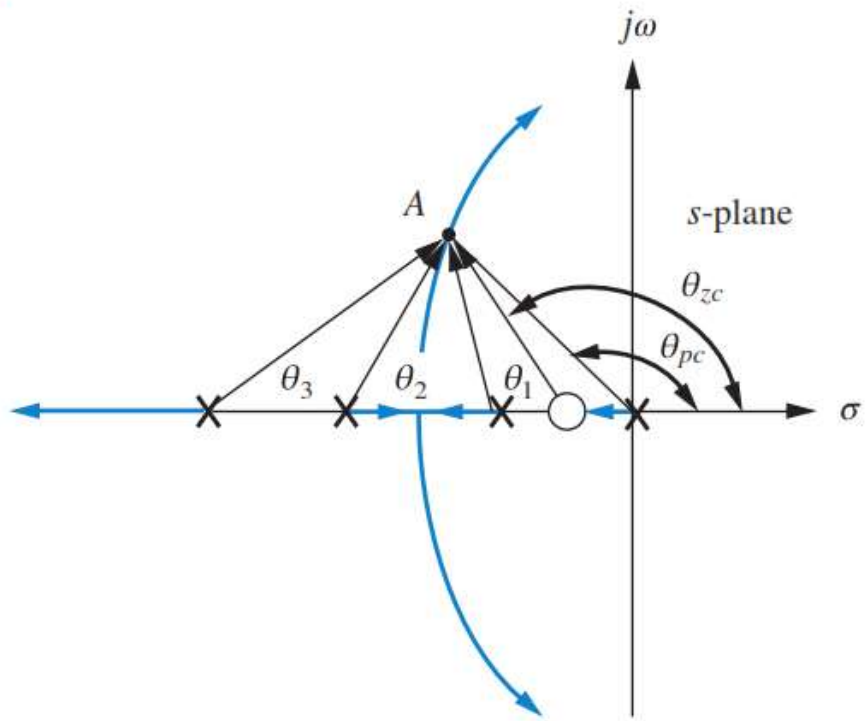
Lecture: Design via Root Locus

Improve steady-state error and transient response	PID	$K\dfrac{(s+z_{\text{lag}})(s+z_{\text{lead}})}{s}$	**1.** Lag zero at $-z_{\text{lag}}$ and pole at origin improve steady-state error. **2.** Lead zero at $-z_{\text{lead}}$ improves transient response. **3.** Lag zero at $-z_{\text{lag}}$ is close to, and to the left of, the origin. **4.** Lead zero at $-z_{\text{lead}}$ is selected to put design point on root locus. **5.** Active circuits required to implement. **6.** Can cause noise and saturation; implement with rate feedback or with an additional pole.
Improve steady-state error and transient response	Lag-lead	$K\dfrac{(s+z_{\text{lag}})(s+z_{\text{lead}})}{(s+p_{\text{lag}})(s+p_{\text{lead}})}$	**1.** Lag pole at $-p_{\text{lag}}$ and lag zero at $-z_{\text{lag}}$ are used to improve steady-state error. **2.** Lead pole at $-p_{\text{lead}}$ and lead zero at $-z_{\text{lead}}$ are used to improve transient response. **3.** Lag pole at $-p_{\text{lag}}$ is small and negative. **4.** Lag zero at $-z_{\text{lag}}$ is close to, and to the left of, lag pole at $-p_{\text{lag}}$. **5.** Lead zero at $-z_{\text{lead}}$ and lead pole at $-p_{\text{lead}}$ are selected to put design point on root locus. **6.** Lead pole at $-p_{\text{lead}}$ is more negative than lead zero at $-z_{\text{lead}}$. **7.** Active circuits are not required to implement.

Lecture: Design via Root Locus

Ideal Integral Compensation (PI)





$$-\theta_1 - \theta_2 - \theta_3 - \theta_{pc} + \theta_{zc} \cong (2k+1)180^\circ$$

(c)

A compensator with a pole at the origin and a zero close to the pole is called an *ideal integral compensator*.

Lecture: Design via Root Locus

Ideal Integral Compensation (PI)

$$G_c(s) = K_1 + \frac{K_2}{s} = \frac{K_1\left(s + \frac{K_2}{K_1}\right)}{s}$$

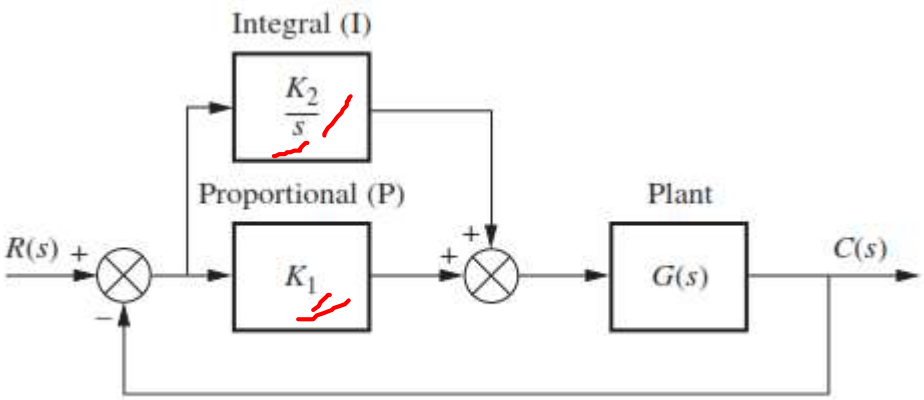


FIGURE 9.8 PI controller

A compensator with a pole at the origin and a zero close to the pole is called an *ideal integral compensator*.

Lecture: Design via Root Locus

Lag Compensation

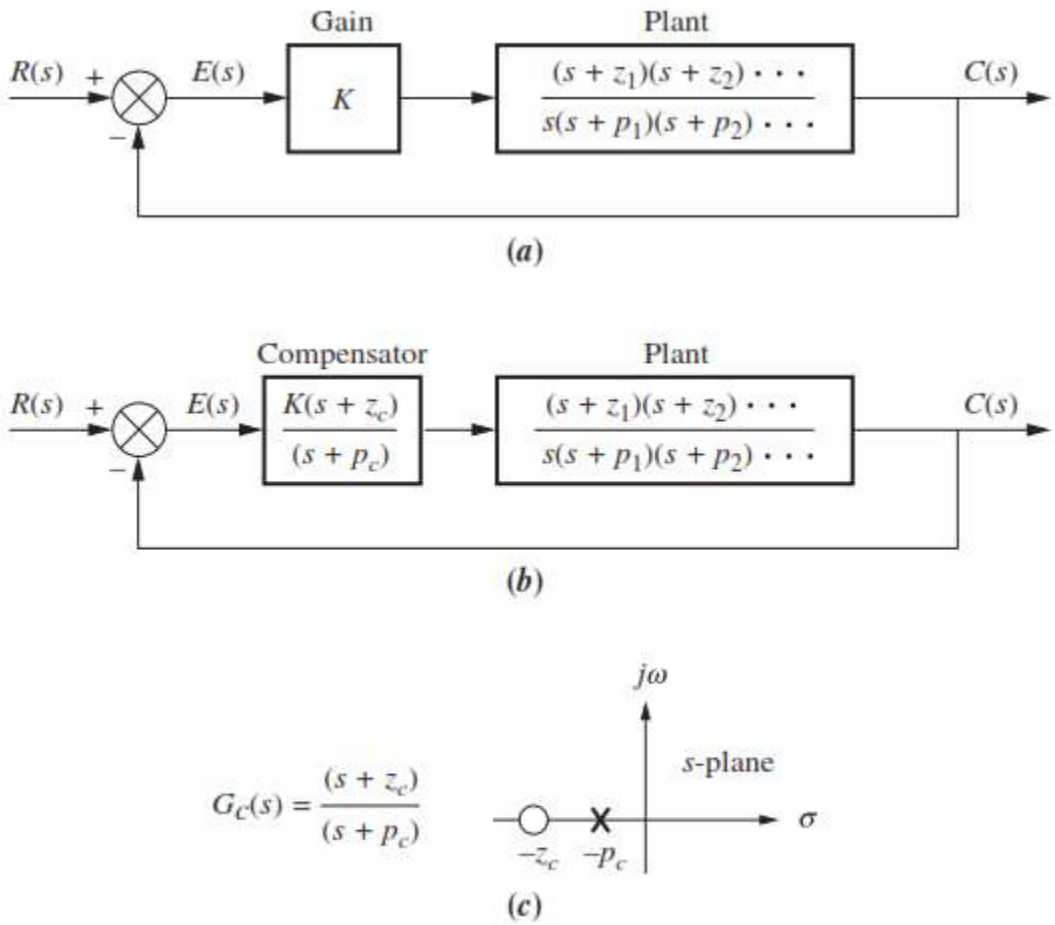
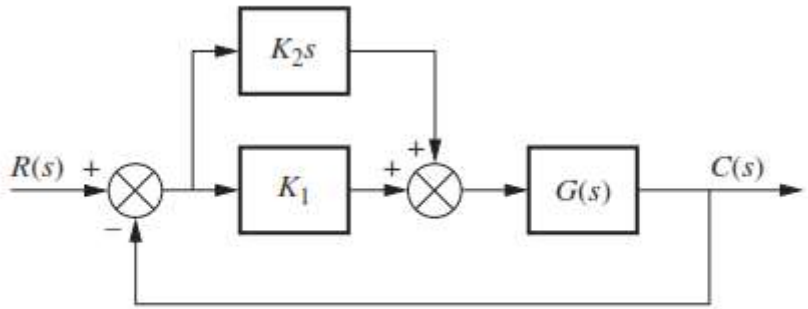


FIGURE 9.9 **a.** Type 1 uncompensated system; **b.** Type 1 compensated system; **c.** compensator pole-zero plot

Lecture: Design via Root Locus

Ideal Derivative Compensation (PD)



$$G_c(s) = K_2 s + K_1 = K_2\left(s + \frac{K_1}{K_2}\right)$$

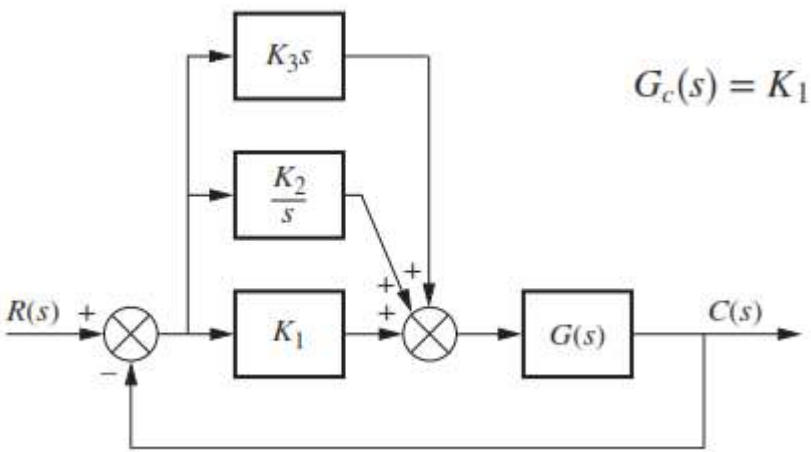
FIGURE 9.23 PD controller

Lecture: Design via Root Locus

Lead Compensation

Lecture: Design via Root Locus

PID Controller Design



$$G_c(s) = K_1 + \frac{K_2}{s} + K_3 s = \frac{K_1 s + K_2 + K_3 s^2}{s} = \frac{K_3\left(s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3}\right)}{s}$$

FIGURE 9.30 PID controller

Lecture: Design via Root Locus

PID Controller Design

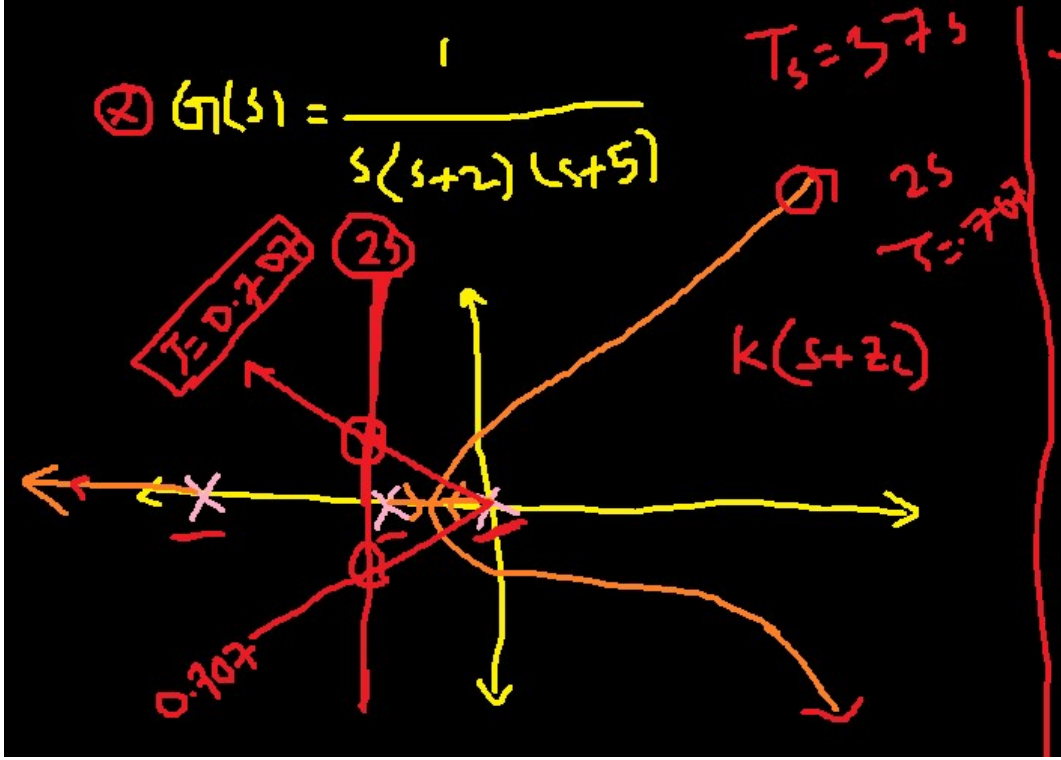
The design technique, which is demonstrated in Example 9.5, consists of the following steps:

1. Evaluate the performance of the uncompensated system to determine how much improvement in transient response is required.
2. Design the PD controller to meet the transient response specifications. The design includes the zero location and the loop gain.
3. Simulate the system to be sure all requirements have been met.
4. Redesign if the simulation shows that requirements have not been met.
5. Design the PI controller to yield the required steady-state error.
6. Determine the gains, K_1, K_2, and K_3, in Figure 9.30.
7. Simulate the system to be sure all requirements have been met.
8. Redesign if simulation shows that requirements have not been met.

Lecture: Design via Root Locus

Lag-Lead Compensator Design

Lecture: Design via Root Locus



Step-01: Uncompensated System Performances

Step-02: Draw Root Rocus

Step-03: Draw the design Criteria

Step-04: Add pole/zero based on your chosen controller

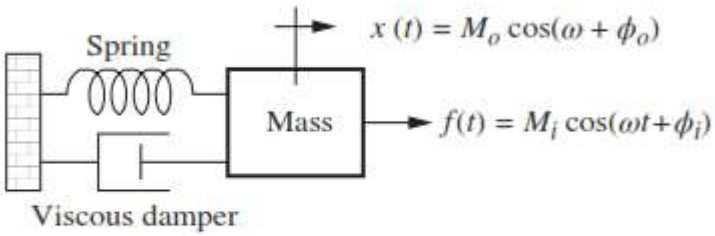
Step-05: Move the added pole/zero to pass the root locus through the desired points.

Step-06: Positioned the closed loop poles at the design point to get the gain.

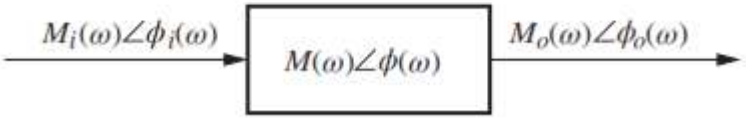
Step-07: Get the controller gain and transfer function

Lecture: Frequency Response

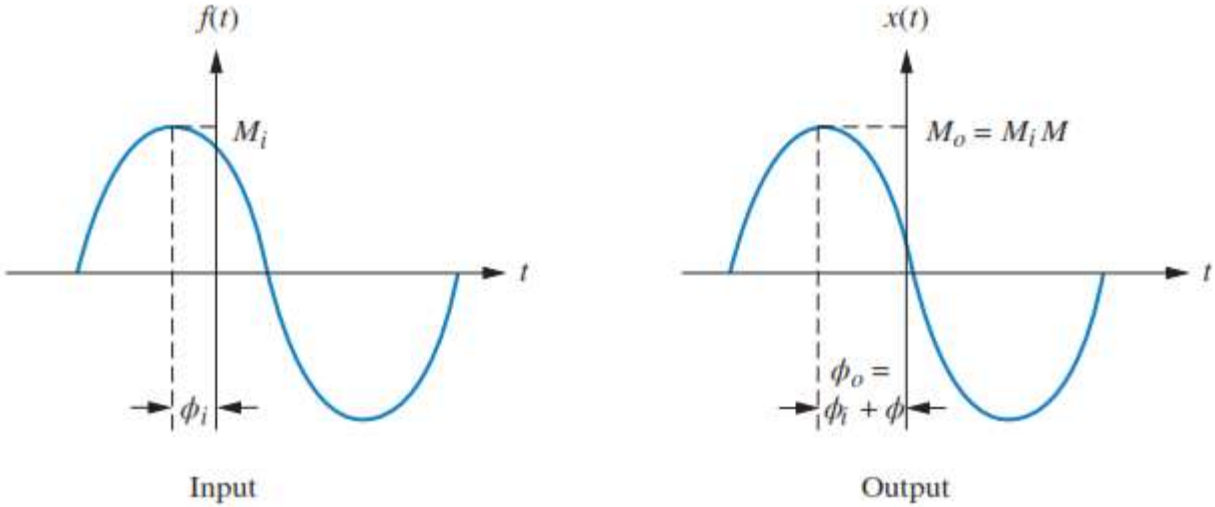
$$M_o(\omega)\angle\phi_o(\omega) = M_i(\omega)M(\omega)\angle[\phi_i(\omega) + \phi(\omega)]$$



(a)



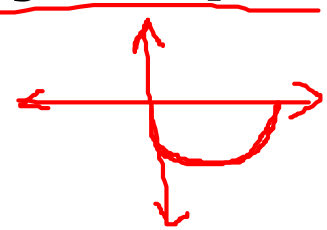
(b)



(c)

Lecture: Frequency Response

$$G(j\omega) = G(s)|_{s \to j\omega}$$



$$G(s) = 1/(s+2)$$

$$|G(j\omega)| = M(\omega) = 1/\sqrt{(\omega^2+4)}$$

$$\phi(\omega) = -\tan^{-1}(\omega/2)$$

$r\angle\phi$

Polar Plots ($M\angle\phi$)

$a+ib$

Bode Plots

(a) Magnitude ($20\log M$ vs $\log\omega$)

dB

(b) Phase (ϕ vs $\log\omega$)

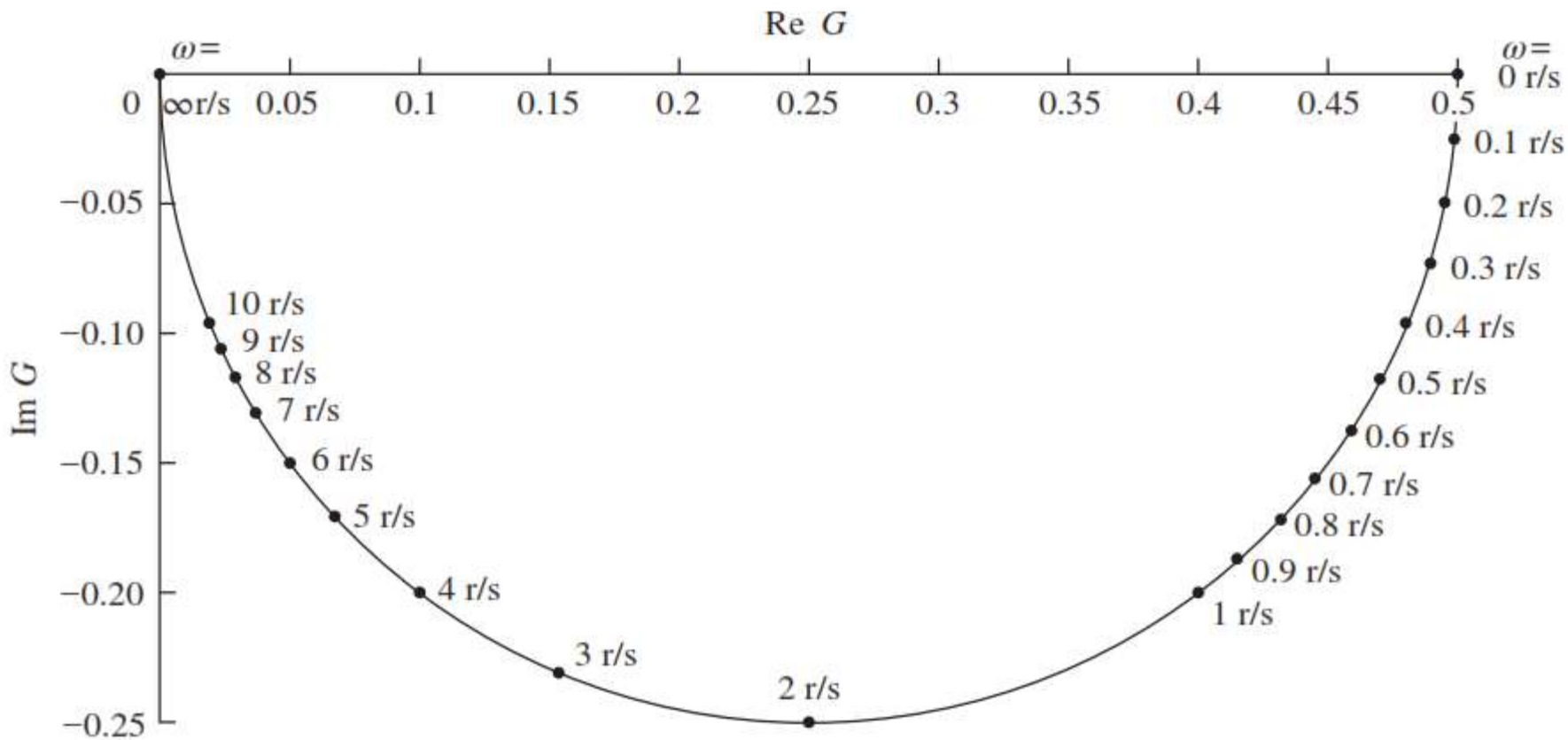
$$20\log M(\omega) = 20\log(1/\sqrt{\omega^2+4}) \text{ vs. } \log\omega,$$

$$\phi(\omega) = -\tan^{-1}(\omega/2) \text{ vs. } \log\omega.$$

$$M(\omega) < \phi(\omega) = 1/\sqrt{\omega^2+4} < -\tan^{-1}(\omega/2) \text{ for different } \omega.$$

Lecture: Frequency Response

$$M(\omega) < \phi(\omega) = 1/\sqrt{\omega^2 + 4} < -\tan^{-1}(\omega/2) \text{ for different } \omega.$$



Note: r/s = rad/s

Lecture: Frequency Response

$G(s) = s + a$

$G(s) = \frac{1}{s+a}$

$G(s) = s$

$G(s) = \frac{1}{s}$

$G(j\omega) = j\omega + a$

$= a\left(\frac{j\omega}{a} + 1\right)$

$M(\omega)$

$\phi(\omega)$

$\omega \to 0$

a

$20\log a$

$\omega \gg a$

$a(j\omega/a)$

$j\omega$

$\omega \angle +90^\circ$

$20\log\omega$

$\frac{1}{j\omega + a}$

$= \frac{1}{a\left(\frac{j\omega}{a} + 1\right)}$

$\frac{1}{a}$

a^{-1}

$-20\log a$

$\frac{1}{j\omega}$

$-j\omega^{-1}$

$\omega^{-1} \angle -90^\circ$

$-20\log\omega$

$j\omega$

$\omega \angle 90^\circ$

$20\log\omega$

$y = 20^{+}$

dB

$+20$

0

$\frac{1}{j\omega}$

$-j\omega^{-1}$

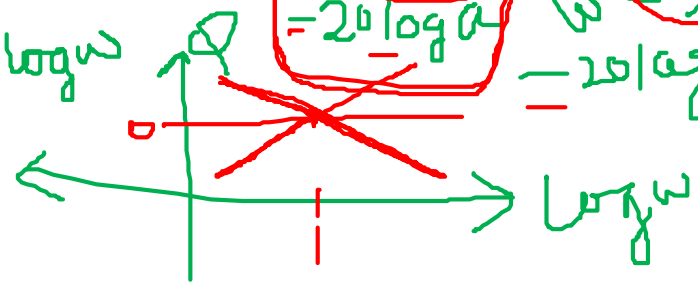
$\omega^{-1} \angle -90^\circ$

$-20\log\omega$

0

$y = mx$

0dB



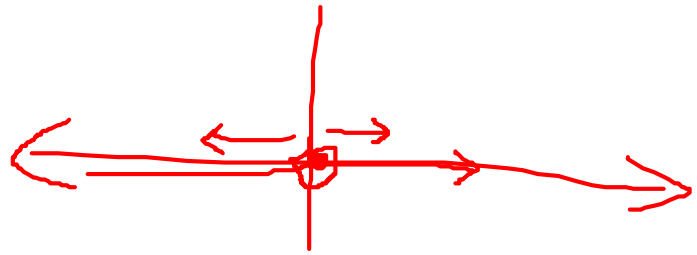
Lecture: Frequency Response

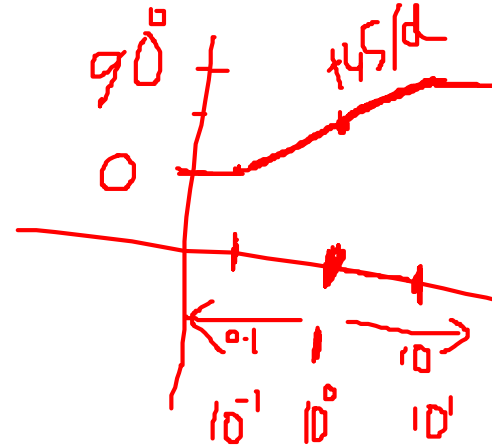
$s+a$

$G(s) = \boxed{s+a}$

$G(j\omega) = j\omega + a$

$= a\left(j\frac{\omega}{a}+1\right)$





1) $\omega \to 0$

$G(j\omega) = a$

$= a\angle 0$

$20\log a$ Vs $\log\omega$

2) $\omega \to \infty$

$G(j\omega) = j\omega$

$= \omega\angle 90°$

$y = 20\log\omega$ Vs $\log\omega = x$

$= 20x$

$20\log 1 = 0$

$20\log 10^1 = 20$

3) Break Fq

$\boxed{\omega = a}$

$G(j\omega) = a(j+1)$

$= \angle 45°$

Lecture: Frequency Response

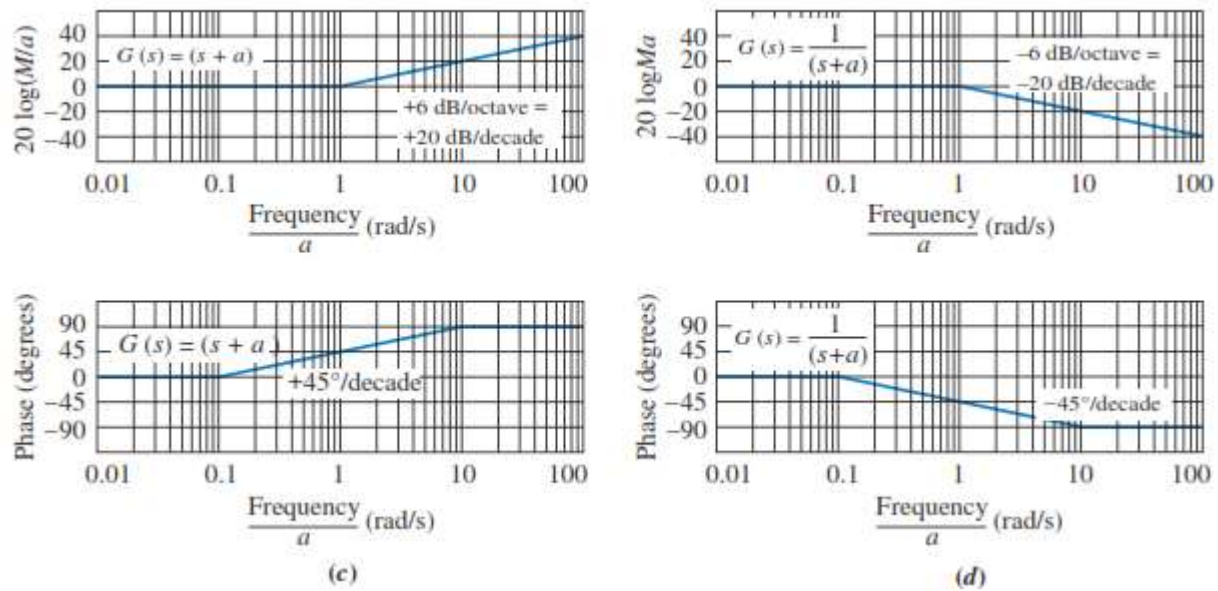


FIGURE 10.9 (*Continued*)
c. $G(s) = (s + a)$;
d. $G(s) = 1/(s + a)$

Figure 10.12

Lecture: Frequency Response

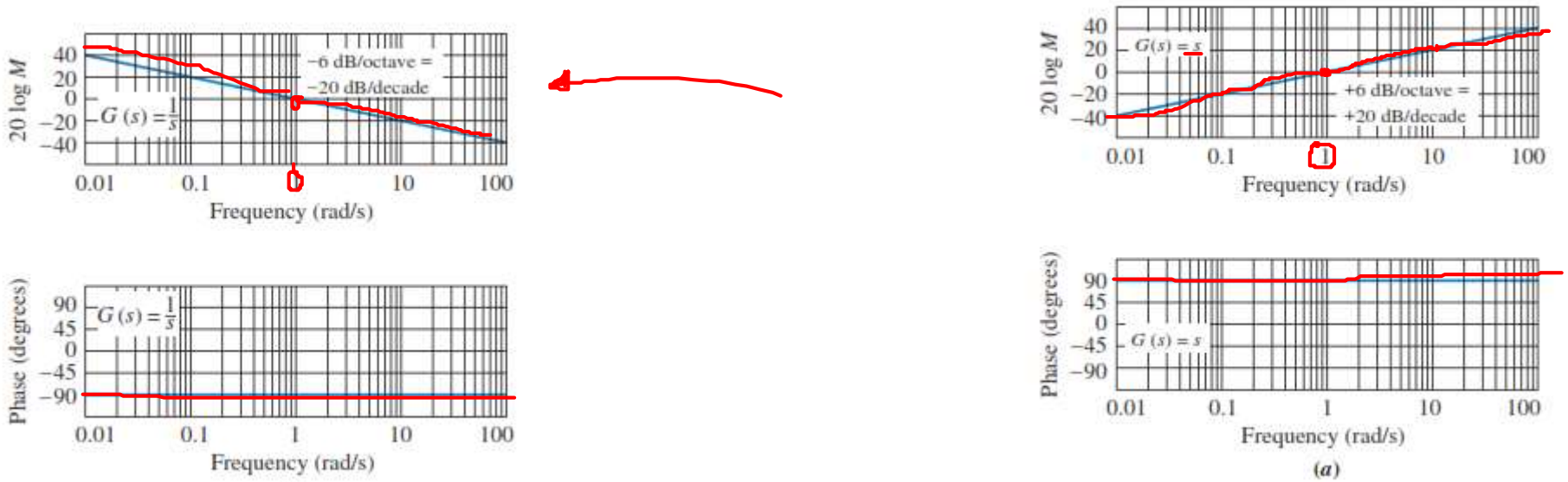
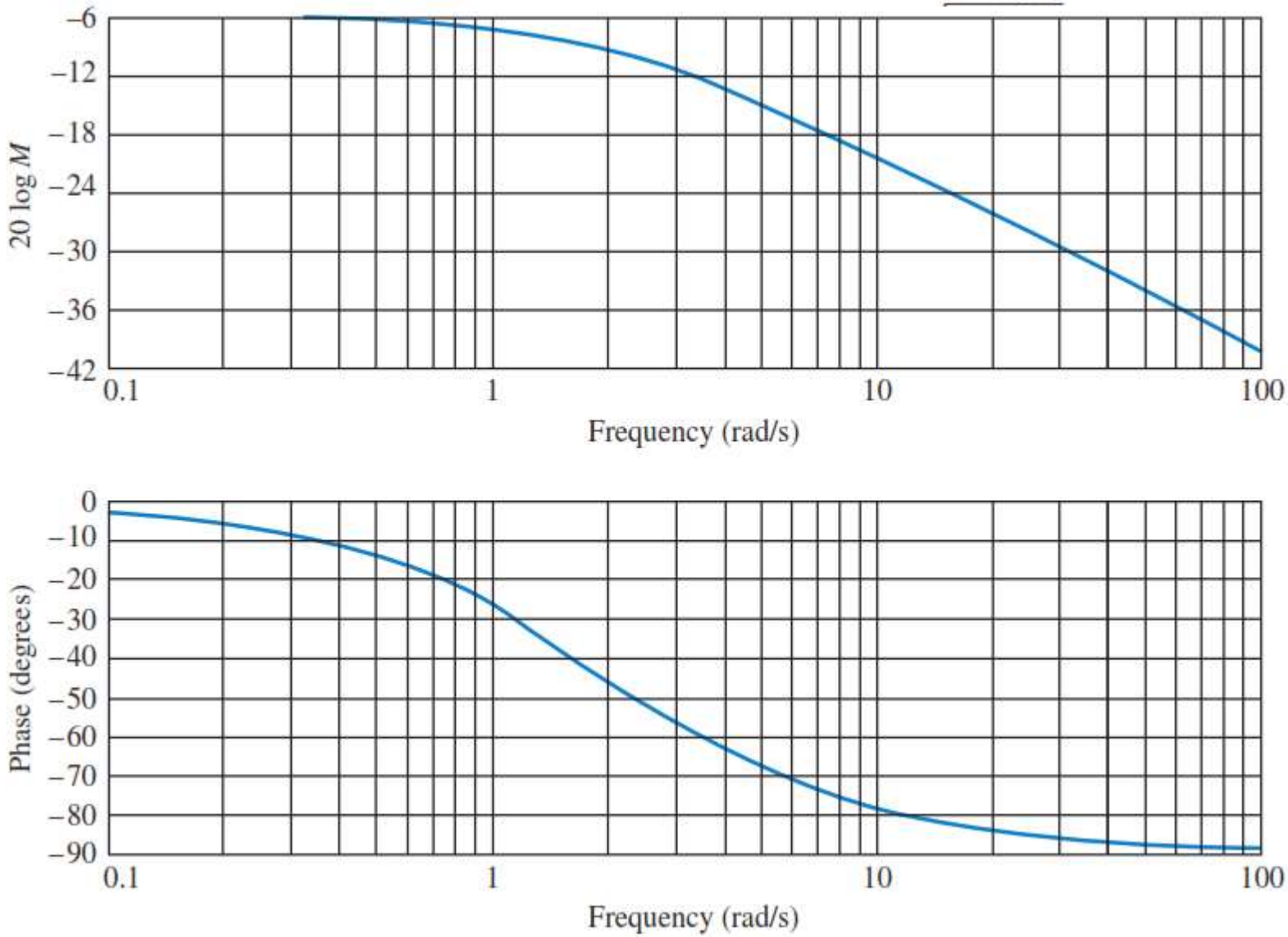


FIGURE 10.9 Normalized and scaled Bode plots for
a. $G(s) = s$;
b. $G(s) = 1/s$;

Lecture: Frequency Res

$$20 \log M(\omega) = 20 \log \left(1/\sqrt{\omega^2 + 4}\right) \text{ vs. } \log \omega,$$

$$\phi(\omega) = -\tan^{-1}(\omega/2) \text{ vs. } \log \omega.$$



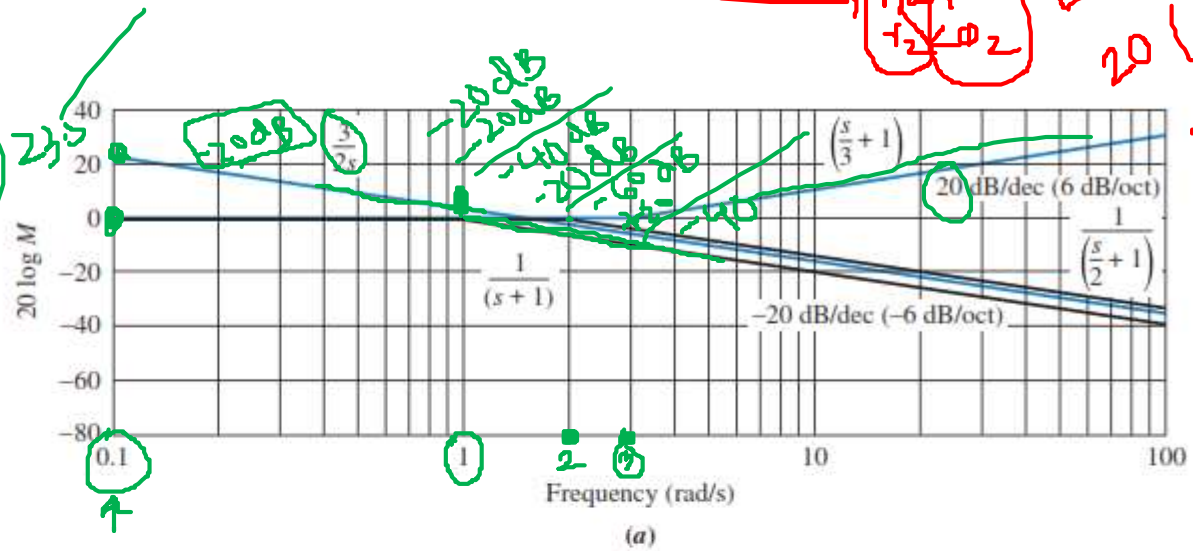
Lecture: Frequency Response

Example 10.2

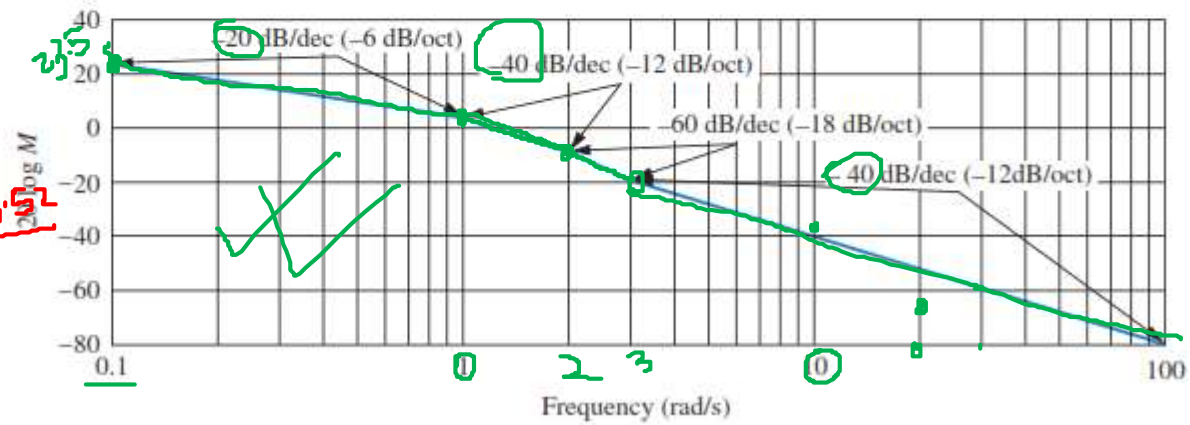
Bode Plots for Ratio of First-Order Factors

PROBLEM: Draw the Bode plots for the system shown in Figure 10.10, where $G(s) = K(s+3)/[s(s+1)(s+2)]$.

Lecture: Frequency Response

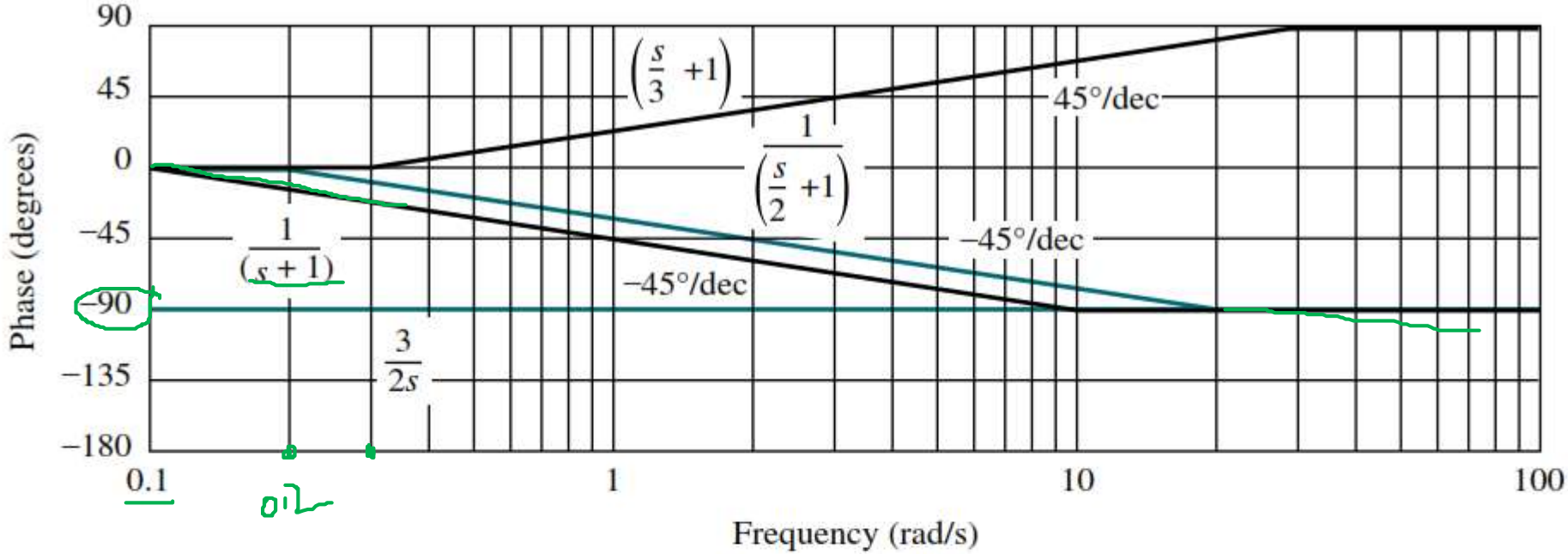


(a)

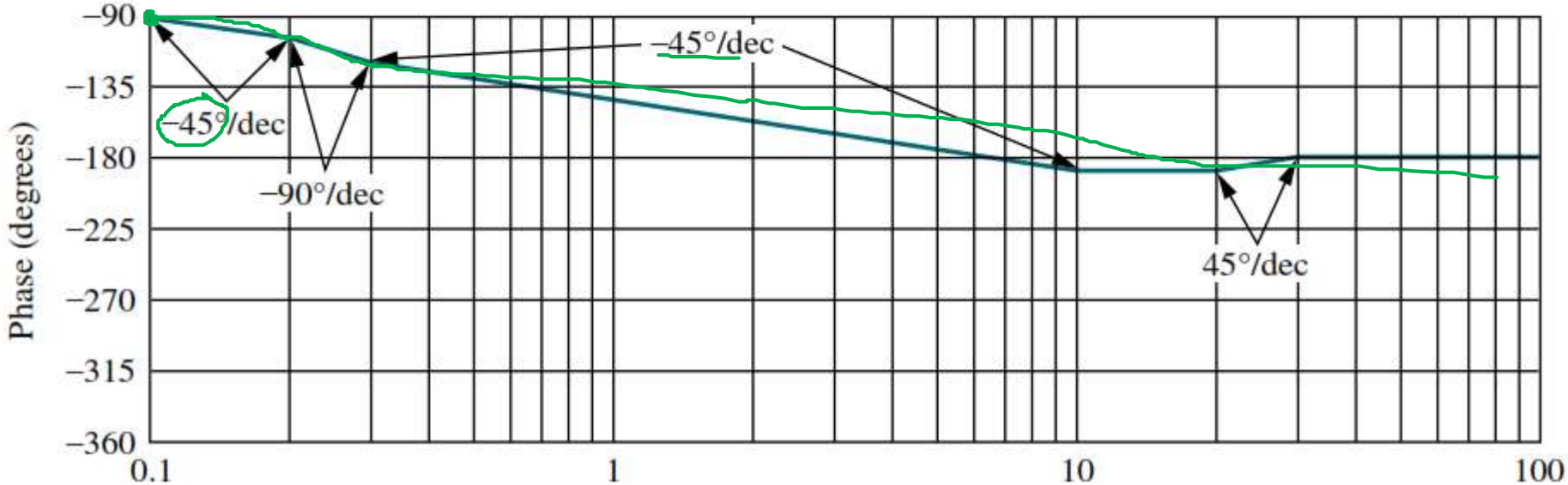


(b)

FIGURE 10.11 Bode log-magnitude plot for Example 10.2:
a. components;
b. composite


90
45
0
−45
−90
−135
−180
Phase (degrees)
$\left(\frac{s}{3}+1\right)$
45°/dec
$\frac{1}{\left(\frac{s}{2}+1\right)}$
−45°/dec
$\frac{1}{(s+1)}$
−45°/dec
$\frac{3}{2s}$
0.1
1
10
100
Frequency (rad/s)

(*a*)


−90
−135
−180
−225
−270
−315
−360
Phase (degrees)
−45°/dec
−45°/dec
−90°/dec
45°/dec
0.1
1
10
100